

Do Skilled Migrants Compete with Native Workers?

Analysis of a Selective Immigration Policy

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Abstract

Skilled migration is seen as a solution against skill shortages, but its impact on native workers facing increased competition remains debated. To investigate the longer-term potential for displacement, this paper exploits a 2008 reform in France facilitating the hiring of foreign workers within a list of tight occupations. The analysis relies on administrative data and applies a difference-in-differences approach. Results show that the reform successfully boosted migrant hires without affecting native employment. In addition, the negative pressure on occupational wages is much larger among migrants than natives, suggesting that such policies can be a viable solution in the long run.

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1 Introduction

Several Western countries have adopted selective immigration policies to attract foreign workers with valuable skills, but there is still a lack of consensus regarding their effectiveness and their consequences for incumbent workers.¹ In the COVID-19 aftermath, this question has become even more central in the policy debate, following the emergence of severe shortages in several key occupations combined with tighter immigration restrictions in countries such as the UK and the US. If attracting migrants in occupations suffering from shortages significantly degrades the salary conditions of natives in these jobs, it is likely that the shortages will persist in the long run, by discouraging locals from investing in these skills. If, on the contrary, native workers in target jobs are largely shielded from the negative pressure on wages, it suggests that these policies can be a viable solution.

Despite the canonical model's prediction that a positive immigrant supply shock should depress the price of labor (Borjas, 2003), the empirical literature has often failed to find large negative wage effects.² This paper exploits a French reform relaxing immigration constraints within a list of occupations facing skill shortages. By precisely identifying affected workers by their professions, it tests the model predictions in a context where most attenuation channels highlighted in the empirical literature are inactive. Utilizing firm-level data, it examines intra- and inter-establishment transmission mechanisms, highlighting the key role played by firms in shaping the impact of supply shocks on wages. The main objects of interest in this paper are the wage conditions that migrants and natives face in the occupations targeted by the reform, which are informative of the long term efficiency of selective policies in addressing skill shortages.³

The policy under study is adopted in France in January 2008, lowering the firms' administrative

¹For instance, Canada and Australia have a visa point system that encourages high-skill immigration, the H-1B visa in the US aims to attract skilled immigrants in tight occupations, and the UK favors immigrant workers possessing qualifications that are in shortage domestically.

²Some reasons highlighted are skill downgrading (Dustmann, Frattini and Preston, 2012), differential sorting across firms (Arellano-Bover and San, 2020; Brinatti and Morales, 2021), and natives switching to better jobs (Peri and Sparber, 2009; Foged and Peri, 2016).

³This approach differs from the one taken by Foged and Peri (2016) for instance, who follow affected workers along their careers and across occupational switches.

costs for hiring extra-European workers within a list of 30 tight occupations. Its primary aim is to help employers access the skills that they need. The professions concerned mostly require a technical specialization from upper secondary or tertiary education, and can thus be considered middle to highly skilled. One advantage of this reform is that it introduces a second list of occupations selected based on the same criteria, but where the administrative procedure is only lessened for workers from Romania and Bulgaria.⁴ This provides a natural control group for a difference-in-differences estimation, under the assumption that occupations in both lists were evolving comparably before the reform, and that the ones in the treated list are exposed to a larger inflow of workers after the policy change. The analysis is based on two administrative employer-employee datasets recording new hires, employment and wages by detailed occupation and broad nationality, which enables the study of the differential impact of the reform on natives and migrants. Finally, the analysis is carried out within cells of occupations $\times$ establishments, which makes it possible to disentangle within firm mechanisms from sorting across firms.

Results show that the policy increases the flows of migrant hires by 51% and the stock of migrant employment by 21% on average in the six years following the reform. This indicates that employers took advantage of the lighter regulations by employing more foreign workers. Native employment is unaffected in the short run, and towards the end of the observation period, five years after the reform, it even shows a positive trend. The latter can be due to firms reacting positively to the reduction in hiring costs (Horton, 2017), or to firms expanding thanks to the reduction of their skill constraints (Clemens and Lewis, 2022; Signorelli, 2020). These findings are a first indication that the policy was effective in reducing shortages. Nonetheless, if salary conditions in these jobs are significantly deteriorated for natives, shortages might reemerge in the future. Overall, I find that wages decrease in reform occupations following the policy change, but do so by a much larger extent for migrants (-1.8%) than for natives (-1.3%). This difference might however be partly due to changes in the composition of newly arrived migrants. When I focus on workers already active in France prior to the reform, I find a smaller but still significant negative effect on wages, and the difference in the impact for migrants and natives persists (-0.9% vs -0.5%). To get a better sense

⁴Bulgaria and Romania are European Union countries under a transitory regime that temporarily prevents them from accessing the common labor market freely.

of the magnitude of the effect, I derive the supply elasticity of wages: a 1% increase in migrant employment decreases incumbent migrant wages by 0.07% and incumbent native wages by 0.04%. The elasticities to total changes in employment are larger, around -0.25 for incumbent migrants and -0.11 for incumbent natives. All in all, it appears that natives in affected occupations are largely shielded from the additional competition generated by migrants, supporting the claim that this kind of policies can be effective in reducing shortages in the long run.⁵

Compositional changes are unlikely to explain the differential wage effect between migrants and natives. In addition, natives tend to sort in smaller establishments relative to migrants. Controlling for differential sorting across establishment sizes thus exacerbates rather than explaining the difference in the wage impact. It is thus likely that the latter is explained by within firm mechanisms. I make use of the novel framework put forward by [Amior and Manning \(2020\)](#) to provide suggestive evidence that bargaining power and imperfect degrees of substitution contribute to explaining the differential effect on native and migrant wages.

This paper contributes to the literature on the labor market effects of high-skill immigration, which has typically been studied in the context of H-1B visas in the US.⁶ By focusing on the effect of migration in occupations facing a direct increase in migrant employment, it distinguishes itself from studies on broader supply shocks affecting entire areas. It can thus isolate the impact on the portion of the labor market that is the most likely to suffer following the inflow of migrants, abstracting from aggregate mechanisms contributing to attenuate such effect.⁷ Only a handful of papers explore the effect of occupation-specific migration shocks. [Friedberg \(2001\)](#) looks at the massive inflow of Russian Jews into Israel in the early 1990s and finds no effect on native wages, [Bratsberg and Raaum \(2012\)](#) look at differential exposure to immigration across different

⁵It is important to notice that these results do not imply that the group of affected workers, defined as people employed in targeted occupations at the moment of the reform, lose out overall. In fact, I show that if workers are followed along their careers, their wages are either not affected or even react positively, as already shown by [Foged and Peri \(2016\)](#).

⁶See [Doran, Gelber and Isen \(2022\)](#), [Kerr and Lincoln \(2010\)](#), [Kerr, Kerr and Lincoln \(2015\)](#), [Mayda et al. \(2023\)](#), [Mayda et al. \(2018\)](#), and [Peri, Shih and Sparber \(2015a\)](#) for studies on the US. See [Beerli et al. \(2021\)](#) for a study on Switzerland.

⁷For instance, the expansion of industries and firms using these skills abundantly ([Dustmann and Glitz, 2015](#)), and the growth in city employment following productivity boosts ([Peri, Shih and Sparber, 2015b](#)). While these mechanisms are interesting from a policy perspective, they attenuate the negative wage pressure generated by competition, which is what I want to capture in this paper.

sub-segments of the Norwegian construction sector and find a native wage-supply elasticity of 0.06, which is about two thirds of the magnitude that I find in this paper, and [Kuosmanen and Meriläinen \(2022\)](#) look at Finnish wages in the construction sector after the EU enlargement, finding large negative effects. While the main focus of these papers is the effect on native workers, the main objective here is to quantify and explain differences in the effect between natives and migrants. Finally, [Peri and Sparber \(2009\)](#) and [Foged and Peri \(2016\)](#) focus on the effect of migration for exposed natives, including those that switch occupations. In this paper, I am interested in capturing the changing wage conditions for natives and migrants that remain in target occupations, in order to capture the emergence of disincentives to specialize in these skills that could make the shortages persist in the long run.

This paper is among the firsts to disentangle mechanisms taking place within- and between firms, contributing to the understanding of the relative importance of these different margins of adjustment.⁸ In the estimation of the within-firm channels, I follow the novel framework proposed by [Amior and Manning \(2020\)](#), which revisits the estimation of the elasticity of substitution between migrants and natives by relaxing the assumption of perfect competition in the labor market ([Card, 2009](#); [Manacorda, Manning and Wadsworth, 2012](#); [Ottaviano and Peri, 2012](#)). My results highlight the importance of considering monopsonistic competition and differential degrees of bargaining power when studying the effects of migration shocks.⁹

From a policy perspective, addressing skill shortages is becoming a pressing issue for countries willing to keep up with the technological frontier. [Raux \(2019\)](#) shows that US firms applying for H-1B visas do so because they face hiring difficulties, and [Mayda et al. \(2023\)](#) estimate that the

⁸A very recent literature has started to explore the role of sorting across firms in determining the impact of migration ([Arellano-Boyer and San, 2020](#); [Orefice and Peri, 2020](#); [Brinatti and Morales, 2021](#)), but these works have not explored mechanisms taking place within firms. [Doran, Gelber and Isen \(2022\)](#) use information from the H-1B visa lotteries at the firm level, and show that obtaining one additional migrant generates some crowding out of native employment in the firm. This paper differs from their analysis by looking at the overall impact of increased access to migrants, instead of the marginal one. Finally, the French reform under study is more targeted on occupations facing shortages than the American H-1B visa system. The higher selectivity reduces the scope for employment crowding-out, putting additional weight on the wage effect.

⁹This channel has been incorporated in search and matching models where natives and migrants have different reservation wages ([Chassamboulli and Peri, 2020](#); [Chassamboulli and Palivos, 2013](#); [Nanos and Schluter, 2014](#); [Moreno-Galbis and Tritah, 2016](#)), but has been much less studied in the context of natural experiments in the migration literature. A notable exception is the paper by [Naidu, Nyarko and Wang \(2016\)](#).

rationing of H-1B visas in recent years reduced employment, sales and profits of US firms. This paper reveals that, even in the context of the common European labor market, which counted about 220 million workers in 2007,¹⁰ businesses can still suffer from skill shortages. This is in line with the low degree of cross-country mobility highlighted by [Dorn and Zweimüller \(2021\)](#). In a companion paper, I show that the firms that were the most constrained by the lack of skills are able to grow faster thanks to the policy change ([Signorelli, 2020](#)). Selective immigration policies seem thus to be a viable solution to reduce skill shortages, while the burden imposed on natives with similar skills is rather limited.

The rest of the paper is organized as follows. Section 2 describes the reform, Section 3 presents the data, Section 4 discusses the empirical strategy, Section 5 details the main results, Section 6 explores the channels behind the main findings, and Section 7 concludes.

2 The Reform

In France, the labor law gives priority to current residents and EU nationals in the hiring process. For all occupations in the labor market, prior to the reform of 2008, if an employer wished to hire a non-European citizen that did not currently reside on French territory, he had to apply for a work authorization from the regional Office of Foreign Labor (*Service de Main d'Oeuvre Etrangère* - SMOE) that was only granted if two conditions were met. First, the employer had to prove to have searched extensively for a priority candidate before considering hiring a non-European worker.¹¹ Second, the occupation under question had to appear as "tight" in the statistics collected by the French Employment Office.¹²

¹⁰Figure taken from Eurostat: number of workers in the European Union (EU27).

¹¹In practice, the employer needed to transmit the posted job offer to the SMOE, as well as, if applicable, the CVs of all candidates who responded to the offer and a letter stating the reasons for the company refusal. While officially the employer needed to prove that the offer remained vacant for at least three weeks, in practice, the length of time imposed on employers by the some regional SMOE offices could be as long as two months, depending on the region and the type of contract.

¹²The tightness indicator measures the ratio between available vacancies in each occupation and the pool of unemployed workers possessing the required competencies for the job. In practice, the regional SMOE office consults the local French employment office and together they assessed the severity of the shortage in the occupation in question.

Figure A1 in Appendix reports a diagram adapted from OECD (2017) that summarizes all the administrative steps that needed to be undertaken by an employer before hiring a migrant worker. This procedure was lengthy and burdensome for firms. All demands had to be submitted in paper format, and despite the fact that the official time to process requests was set to a maximum of 2 months, it often took much longer in practice. Given the complexity of the procedure, France is characterized by the fact that the majority of foreign nationals in possession of an economic visa were already present in the country as students, while their recruitment at the end of their university curriculum granted them a change of visa status.¹³

In January 2008, the French government introduced a new legislative decree facilitating the hiring of extra-European workers within a list of 30 occupations characterized by a high level of tightness in the labor market. The jobs in the list account for about 10% of the French private sector employment. The new law states that, for the occupations concerned, the employer is no longer required to prove the prior search for a priority candidate but is automatically granted the authorization to hire a migrant worker. The aim of the reform is to help firms recruiting in skills that suffer from a scarcity of domestic labor. Suggestive evidence reveals that these changes significantly reduced the time and the amount of uncertainty that firms face when trying to hire a foreign worker.¹⁴ To avoid abuses, the Office of Foreign Labor carefully checks that the qualifications and work experience of the candidate match the occupation for which he is hired and that the contract conditions, notably in terms of salary, are in line with the legal minima for the position.¹⁵ This reduces the possibilities for firms to mischaracterize vacancies in order to hire foreign workers more easily, and ensures that the migrants hired through the new policy actually perform the tasks corresponding to their stated occupation. The office of foreign labor

¹³A report from the French Ministry of Interior states that in 2005 60% of the recipients of their first economic visa were already present in the country (mostly as students). This proportion went down to 50% in 2012 ("Les données de l'immigration professionnelle et étudiante", French Ministry of Interior).

¹⁴In terms of time, the employers no longer need to prove that their vacancy was unfilled for at least three weeks until up to two months. In terms of uncertainty, the simplification of the procedure restricts the room for arbitrary refusals expressed when all the conditions are met. A report from the French Ministry of Interior reveals that in 2012 30% of economic visa applications were refused in jobs where all the restrictions applied, while only 4.4% of the applications were refused in jobs where the two above-mentioned restrictions were lifted ("Les données de l'immigration professionnelle et étudiante", French Ministry of Interior).

¹⁵In France, in addition to a national minimum wage, different sectors negotiate specific minimum wages that are set above the national one. The verification of diplomas is simplified by the fact that a large portion of newly hired migrants completed their education in France.

re-evaluates the visa every year and only grants it if conditions are met. While reducing the likelihood of frauds, the regulation leaves a lot of flexibility in terms of wage setting, since the occupations in the list are generally paid well above the minimum standards.

In a first step, the list is defined at the national level, and in a second phase, each region selects a subset of occupations that remain tight at the local level.¹⁶ Table A1 in the Appendix reports the full list of target occupations and details the number of regions to which each of them applies. The main economic activities concerned are computer science (2 occupations), construction (4 occupations), electricity and electronics (4 occupations), and mechanical construction and metal processing (4 occupations). The main professional categories involved are technicians, engineers and foremen.¹⁷ Most of these occupations require some technical specialization from upper secondary or tertiary education, and can thus be considered as middle- to high-skilled jobs.

For the identification strategy, I take advantage of the fact that the legal change under study is part of a larger effort to reform France’s economic migration policy. In particular, the working group in charge of the reform established at the same time an extended list of 150 occupations to be open without restrictions to European nationals coming from member states under transitory regimes.¹⁸ The latter includes all 30 occupations open to non-EU citizens and 120 additional ones. The definition of both lists is based on the same set of tightness criteria, with the idea that the tightest occupations should enter the first and the slightly less tight the second. The professions open only to the EU States under transitory regimes thus provide a natural control group for those open to all nationalities, under the assumption that, while they are similar in terms of pre-reform characteristics once we control for the probability of entering the first list based on tightness, the treatment intensity is much lower in the first than in the second.¹⁹

¹⁶Only five jobs apply to the entire French territory, while the others are only valid in certain areas.

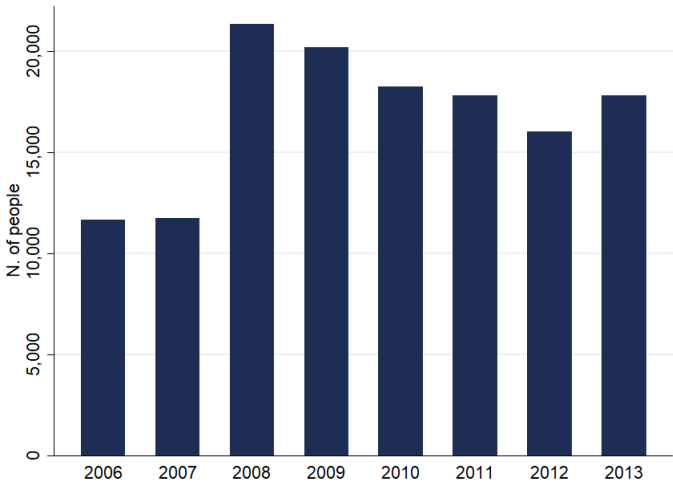
¹⁷Figure A2 and A3 in Appendix detail the exposure to the reform across broad categories of occupations and sectors.

¹⁸At the time of the implementation of the reform this concerned only Romanian and Bulgarian nationals, which joined the European Union in 2007, but later it also included Croatians (EU members since 2013). Workers from these countries did not immediately obtain the right to work in all member states, and in France they continued to be subject to the same labor market restrictions as extra-Europeans during a probation period that lasted until 2014.

¹⁹To precisely measure tightness in each occupation, the working group considered several indicators collected quarterly by the Employment Office: i) the ratio between job supply (vacancies) and demand (unemployed with relevant skills), ii) the volume of job supply, iii) the volume of job demand, iv) the evolution in the stock of demand

Despite the availability of a precise description of the indicators used to define both lists, it was impossible to find a threshold in the tightness data determining the inclusion of an occupation in the reform, such that among severely tight jobs we find at the same time occupations in the treated list, occupations in the EU extended list, and some occupations that are in none of the two. This is due to the fact that the final groupings were decided after the incorporation of some additional dimensions. First, the findings from a report predicting the long-term nature of the current occupational shortages,²⁰ and second, the outcome of negotiations with the social partners that took place at the end of October 2007 (OECD, 2017).²¹ While this rules out the option of a regression discontinuity design, it increases the degree of comparability between the treated and control groups in a difference-in-differences setting. Additionally, the fact that the final decision was only taken at the very end of 2007 rules out the presence of anticipation effects.

Figure 1: Annual Flows of Economic Migrants



Source: National statistics on immigration published by MI-DGEF-DSED.

Figure 1 provides the first descriptive evidence that the labor market did react to the policy

and supply, v) the turnover rate of job-seekers at the end of the month, and vi) the share of long-term contracts within the job offers. All the indicators are collected periodically for each of the 22 regions of metropolitan France and for 225 categories of occupations.

²⁰Report of the *Conseil d'analyse stratégique* (CAS) and of the *Direction de l'animation de la recherche, des études et des statistiques* (DARES) called « The Occupations in 2015 », which came out in 2007 and determined the future skill needs of different sectors.

²¹Unfortunately, there is no detailed information available on the content of such negotiations, except from the fact that the skill level of occupations was taken into account such that lower skilled occupations needed a higher level of tightness to be included. What is important for identification is that the decision to include an occupation in one or the other list is uncorrelated with underlining trends in outcomes. In the analysis I provide several tests for this assumption.

change by attracting larger inflows of migrant workers. According to the aggregate statistics published by the Ministry of Interior, the number of economic visas delivered every year by the French government nearly doubles between 2007 and 2008, jumping from 12 thousands to more than 20 thousands in only one year, and then remaining stable at a higher level, despite the hit of the economic crisis in 2009. In contrast, all other categories of immigration do not show any significant change in size around the same years, as shown in Appendix Figure A4. These patterns are consistent with the expected effect of the reform, since the only migrants directly impacted are those that apply for economic visas in order to work for French firms.

3 Data

The main administrative dataset used in the analysis is the annual payroll tax records (*DADS postes*), which contains information on the universe of employment contracts within the French private sector. In particular, it records information on the number of hours worked in the year, the gross and net salary perceived, the occupation code, and some demographic characteristics of the worker including the age, gender, and whether the worker was born in France or abroad.²² It further includes identifiers for the employing firm and establishment, which can be followed over time, and records their sector of activity and regional location. While this data provides a panel dimension at the firm and establishment level, workers' identifiers are not consistent from one year to the next, and therefore cannot be tracked. Nonetheless, each yearly database includes information on the situation of the individual in the previous year, including the employing firm and establishment identifier, occupation, wage and hours worked at $t-1$. A recent paper by Babet, Godechot and Palladino (2022) shows that it is possible to use the overlapping information over two consecutive years to match workers between yearly files, and to obtain a single match for 98% of the individuals. I follow the codes made available on the authors' website to construct

²²The country of birth indicator is a proxy for the population directly affected by the reform, given that some foreign born are permanent residents in France or naturalized, and are thus allowed to work anywhere without restrictions. Nonetheless, given that the discontinuous change in migration flows in 2008 only concerns economic visas (Appendix Figure A4), it seems reasonable to assume that the identification strategy will detect changes in economic immigration flows driven by the reform.

individual worker identifiers that can be followed over time for the quasi-entirety of the *DADS postes*. I also rely on their code to correct for measurement error in the country of birth variable. Appendix B provides additional details on the data processing.

I restrict the analysis to the French private sector without agriculture. Prior to 2008, only establishments with more than 20 employees are required to report the detailed occupation code of each employee, while smaller establishments report it on a voluntary basis.²³ To avoid potential biases linked to voluntary reporting, I exclude from the sample plants with less than 20 employees. To avoid capturing endogenous changes of sector and/or location, I assign these attributes based on those reported in the first year of establishment appearance in the data, and keep them fixed over time. Finally, to avoid capturing endogenous reclassification of the occupation of individuals within establishments, I fix them based on the first occupation reported by the individual when joining the plant.

The final dataset used in the analysis is at the level of establishments $\times$ occupations that are followed over time, since the main objectives of the analysis are i) to evaluate how the supply shock generated by the reform modified the conditions within the occupations of the list; and ii) to delve into firms' labor demand to understand the mechanisms behind the effects. Employment is thus summed within cells, while the hourly wages are averaged within cells, separately for natives and migrants. The main outcomes of interests are the following :

- Total employment stock of migrants and natives in the establishment $\times$ occupation in a given year, expressed in terms of full-time equivalent.
- Average hourly wages of migrants and natives in the establishment $\times$ occupation in a given year.
- Average hourly wages of incumbent migrants and natives in the establishment $\times$ occupation in a given year, defined as the wages of individuals that were already working in France before 2008.

²³Occupation codes are reported at the most disaggregated level of the French PCS classification: 4-digits.

- Average hourly wages of migrants and natives in the establishment $\times$ occupation in a given year, residualized of individual characteristics including age, age squared, gender, and years of experience within the panel.²⁴

The average wage of migrants and natives captures the evolution of the overall conditions in the occupation for each type of worker. The average wages of incumbent workers capture the effect of the reform on the occupational conditions net of composition changes introduced by the newly arrived workers, but also excluding the effect on young workers that just entered the labor force. Finally, the average wage residualized of individual characteristics aims at correcting for composition effects without excluding from the sample recent graduates. In the analysis, I present the results obtained on each of these outcomes.

To test the robustness of the effect on employment I also look at the impact of the reform on hiring flows measured using a different administrative database. The data on hiring flows used for this analysis is derived from two sources: the monthly declaration of labor flows (*Déclarations des Mouvements de Main d'Oeuvre* - DMMO), which is collected for all establishments with more than 50 employees, and the quarterly survey of labor flows (*Enquête sur les mouvements de main d'oeuvre* - EMMO), which is collected for a representative sample of establishments with less than 50 employees. This database contains information on all hires and separations taking place within establishments, including the detailed occupation, and individual level characteristics, including gender, age and broad country of birth category. Contrary to the *DADS postes*, in this dataset it is possible to distinguish between individuals born within and outside the EU, which allows me to isolate the effect on non-EU migrants.²⁵ On the other hand, I have no information on the contract length of new hires so the hiring outcomes are expressed in terms of headcounts (unweighted). At the establishment level, the data reports the number of employees, the sector and the region where the plant is operating. Establishments can be followed over time but not workers. Here as well I fix the sector and region according to the one of first appearance of the establishment. The

²⁴To obtain the residualized wage information, I estimate an individual level regression with log wage on the left-hand side and demographic characteristics on the right-hand side, including age, age squared, gender, and years of experience in the panel. The regression is estimated separately for native and migrant workers and the residuals are then averaged within the establishment - occupation cells.

²⁵The information on broad nationality categories is directly filled-in by the employers.

main outcomes of interests are the following :

- The number of hires of migrants and natives in the establishment $\times$ occupation in a given year.
- The probability of hiring at least one migrant and one native in the establishment $\times$ occupation in a given year.²⁶

The period considered in the analysis goes from 2004 to 2013, including 4 years prior to the policy change and 6 years after the change.²⁷ Finally, both datasets are merged to a composite index of tightness that varies at the level of occupation $\times$ region $\times$ year and is constructed using the same data used to define the reform. To construct the index, I use all the tightness indicators for the years 2004 to 2007 that were used by the policymakers to decide which occupations to include in the list.²⁸ This database is produced at a quarterly frequency by the French Employment Office (*Pôle Emploi*) and processed by the statistics office attached to the Ministry of Employment (*DARES*).

4 Empirical Strategy

Conceptually, we can think of a tight market as an equilibrium where the demand crosses the supply curve in a portion where the latter is close to being perfectly inelastic.²⁹ Concretely, this

²⁶This is a binary variable equal to 1 if the establishment hires at least one migrant or native in the occupation during the year.

²⁷In 2011 the government tried to reduce the list with the claim that, given the recession, the priority for jobs should be given to French nationals and current residents. However, due to irregularities in the procedure, the change was invalidated one year after, and the list went back to its original composition and was not updated until 2021.

²⁸The index combines the following indicators: i) the ratio between the number of vacancies posted and the number of registered unemployed workers qualified for the job (commonly known as the tightness indicator), ii) the volume of vacancies, iii) the volume of job seekers, iv) the evolution of the stock of vacancies, v) the evolution of the stock of job seekers, vi) the turnover rate of job seekers at the end of the month, and vii) the share of long-term contracts within vacancies. Each indicator is translated into a standardized z-score using the formula $Zscore_i = \frac{x_i - \bar{X}}{\sigma_X}$ and adjusted such that higher values always mean greater tightness, and the final index is computed as the average of the six z-scores. The latter captures all the information used by the commission to define the list of occupations for the reform.

²⁹Somewhat similarly to [Clemens \(2022\)](#), but while in his context the low elasticity of natives is driven by their distaste for the tasks of migrants' occupations, here it is mainly driven by the time required to acquire the skills

corresponds to a context where most of the available local labor with the required qualifications is already employed, while the highly technical content of these occupations makes it difficult to train additional workers in a short period of time, despite the incentives generated by high wages (Saint-Paul and Cahuc, 2009).³⁰

The reform constitutes a positive supply shock, and is thus expected to increase equilibrium employment and to decrease equilibrium wages. In the short term, the increase in employment is expected to be driven by additional migrant hires while native employment will remain mostly unaffected, given the starting context characterized by excess demand. Nonetheless, in the medium term, the effect on native employment could even turn positive, if vacancy posting from firms reacts positively to the reduction in hiring costs (Horton, 2017), or if firms expand thanks to the reform (Clemens and Lewis, 2022). On the contrary, it could turn negative if wages decrease to the point of discouraging native new entrants into these jobs. When it comes to the effect on wages, the negative pressure is expected to be equivalent in magnitude across migrants and natives, if we assume perfect substitutability between these inputs within a given occupation, a competitive labor market, and no changes in the composition of workers in target jobs. These predictions will serve as a benchmark to evaluate the empirical results presented in Section 5. Given that the findings reveal important differences in the impact of the reform on native and migrant wages, Section 6 presents a more complex model of labor demand to unveil the mechanisms behind this result.

The impact of the policy is recovered using a difference-in-differences identification. The treatment group includes the occupations entering the list of the reform at the national level, regardless of whether they apply to the region where the plant is located or not. This can be considered as a measure of treatment exposure, given that not all occupations are "treated" in all regions.³¹ For

necessary for the tasks.

³⁰Descriptive evidence shows that the level of tightness in the occupations of the reform is positively correlated with their average wage, both in cross-section and within occupations across time (Appendix Figure A5), suggesting that this could be an accurate representation of the setting. While native supply will eventually respond to the higher wages, the persistently high level of tightness in these occupations over the period preceding the reform suggests that demand is growing faster than the supply adapts.

³¹This choice is justified by the fact that the regional location of migrants can endogenously change according to where the restrictions are lifted, and multi-plant firms can endogenously adapt the location where they open a vacancy (7% of firms in the data have establishments in more than one region, but they account for 38% of the

the common trend assumption to hold, the control group must include occupations that present labor market conditions that evolve comparably over time, but that are not affected by the reform. Given that the 30 occupations open to non-EU migrants (treated list) are a sub-sample of the extended list applicable to EU states under transitory regimes (EU list), and that both are defined using the same tightness indicators, the remaining occupations of the EU list appear as a suitable control for the analysis. The validity of this control group relies on a series of assumptions. The next paragraphs detail which adjustments to the sample are made to maximize the likelihood that such assumptions are met, they explain which tests are performed to verify their soundness, and they lay out the robustness checks that are provided to ensure that results are insensitive to alternative choices.

First, occupations in both groups need to be subject to similar levels of tightness before the reform, in order for the control to provide a credible counterfactual. The EU list is a natural candidate; given that it was selected based on the same tightness criteria. However, it is much larger and includes some jobs that are much less tight relative to the treated jobs, which might introduce a bias in the results if the tightness differential is a source of asymmetric time-variant changes. To reduce this risk, in the main specification, I exclude control occupations that have a tightness index below the mean observed across all the occupations in the economy and I control for unrestricted time trends in the propensity to enter the treated list based on the tightness criteria used by the policymakers. This can be considered as a "selection on observables" setup. The propensity score is obtained by regressing the treatment indicator on pre-reform tightness and fixed effects for broad job categories.³² To test the comparability of treated and control groups, I verify that pre-trends are parallel and I provide balancing tests for pre-reform characteristics. Further, I compare the main results with those obtained using an alternative control group composed of the occupations in the top two quantiles of tightness distribution within the same occupation

observations since they are larger). In 2007, 85% of employment in treated occupations is actually treated - i.e. is located in a region where the occupation is included in the list. In a robustness check I also present results when treatment is only assigned if the occupation is found in the regional list.

³²The regression is run at the occupation $\times$ region level averaging the composite tightness index on the years 2004 to 2007, and controls for occupation category fixed effects including i) executive and engineering jobs, ii) intermediate professional and technician jobs, iii) and skilled blue collar jobs. I exclude observations outside the common support, which are a small minority (3% of occupations $\times$ region cells are outside the common support).

categories, regardless of whether they appear in the EU extended list, and show that results are highly consistent. Finally, I perform a placebo test where the control group is split into two halves along the tightness dimension, and show that the tighter half does not show different trends after 2008, which suggests that the differences observed in the treated group are driven by the actual effect of the reform.

Second, occupations in the treated and control groups must not be highly substitutable in production. Given that treatment and control occupations can be found within the same establishment, the increased availability of workers in treated occupations would affect the demand for workers in the control group in the presence of such spill-overs, violating the SUTVA assumption.³³ I provide evidence that such production spill-overs are unlikely to be the main drivers of the results by showing that coefficients are robust to excluding from the control group the occupations that are similar in terms of skills required to some occupations in the treatment group - the most likely to be close substitutes.³⁴ Further, I show that results are unchanged when using the alternative control group, while there is no reason to expect that the new controls would be subject to similar production spill-overs. Finally, I show that results remain similar after including establishment $\times$ year fixed effects in the regression, which limits the comparison among treated and control occupations within the same plant, where production spill-overs are expected to be the strongest.³⁵

Finally, the identification relies on the assumption that the financial crisis, which hit France around 2009, had a symmetric effect on treated and control occupations. The main analysis controls for sector $\times$ year fixed effects, which accounts for the potential over-representation of one occupation group in the sectors most affected by the crisis, as it's done by [Bratsberg and Raaum](#)

³³This would introduce an upward bias in the employment results and a bias towards zero in the wage results. I can thus be reassured that the wage coefficients, which constitute the most important findings of the paper, are at worst a lower bound of the true effect. In addition, while the spill-over effects might affect the magnitude of the estimates, there is no reason to believe that they would affect the differential impact of the reform across natives and migrants.

³⁴To define similarity, I consider occupations with the same first three digits out of the four-digits occupation codes. For example, I exclude *Buyers and buyers' support* (control occupation) because too close to *Merchandisers* (treatment occupation), or *R&D technicians in electricity and electronics* (control) because too similar to *design managers in electricity and electronics* (treatment).

³⁵The exclusion of firms that hire both treated and control occupations does not provide a suitable check, because the latter would drop 86% of the observations and 78% of establishments, among which we find the largest firms.

(2012). In support of this hypothesis, I also show in a robustness test that adding establishment $\times$ year fixed effects leaves the majority of the results unchanged, while it is very unlikely that the crisis had a differential effect on different occupations within the same plant, and that results hold when excluding the construction sector, which was the most badly hit by the crisis. Ultimately, the fact that the empirical results reveal a positive effect on employment and a negative effect on wages suggests that they are driven by a differential supply shock rather than a differential demand shock generated by the crisis.

Table A1 in the Appendix reports the list of treated occupations and Table A2 presents the final list of control occupations in the EU extended list. Figure A6 shows the evolution over time of total employment and average wages in the different groups, while Table A3 presents the summary statistics. Finally, Table 1 checks that outcomes are balanced between treatment and control occupations within establishments before the reform. More specifically, it reports the output of a model where outcome variables at the occupation $\times$ establishment level are regressed on the treatment dummy, the propensity score and establishment fixed effects, on the years preceding the reform. Outcomes are inserted both in levels and first differences and the table reports the coefficient on the constant, which reflects the average in the control group conditional on controls, and the coefficient on the treatment dummy, reflecting the difference between the two. Appendix Table A4 does the same exercise for regressions at the region $\times$ occupation level, while Appendix tables A5 and A6 report the same exercise using the alternative control definition. Appendix tables A7 and A8 check for balance in demographic characteristics across treatment and control occupations in regressions at the establishment and regional level respectively.

The summary statistics (Appendix Table A3) show that establishments count on average about 15 native workers and 1.2 to 1.4 migrant workers in each occupation of interest, while they hire about 2 natives and 0.1 migrants every year. Natives are paid on average about 23 gross euros per hour in both treatment and control occupations, while migrants are paid about 23 gross euros per hour in treatment occupations and 21 gross euros per hour in control occupations. The wage levels observed in the occupations of the reform confirm that the jobs in question are relatively skilled, and so are the migrants that are hired to fill them. The evolution of outcomes over time

reveals that, while the levels are different, the trends are highly comparable before 2008 even without adding any control. The balancing tests show that there are some significant differences in levels between treated and control occupations, as for instance higher migrant wages and greater native employment in the treatment group. Nevertheless, there are no significant differences in the changes over time between the treatment and the control groups, comforting the common trend assumption needed for the difference-in-differences analysis.³⁶ At the regional level, there are less differences in the stock variable levels but the decrease in migrant employment is more pronounced within treatment occupations. Similarly, for the demographic variables I do observe some significant differences in levels, especially in the establishment regressions, but much less in trends.

The main analysis starts by checking whether there is a first stage effect of the reform on migrant hires and employment, and evaluates the presence of crowding-in or -out on native employment. In a second step, it looks at how the wages of natives and migrants within affected occupations react to the growth in labor supply. The estimation is based on the following model:

$$Y_{o,i,r,s,t} = \beta_0 + \beta_1 D_o + \beta_2 D_o \times Post_t + \sum_{t=1}^T \delta_t Pscore_{o,r} \times \gamma_t + \gamma_{o,r} + \gamma_{s,t} + \gamma_i + \epsilon_{o,i,r,s,t} \quad (1)$$

where $Y_{o,i,r,s,t}$ captures the outcomes of interest within occupation o , establishment i , region r , sector s , and time t ; the treatment identifier D_o is a dummy that equals one for the occupations in the treatment group and zero for the control group, and $Post_t$ is a dummy equal to one for the years following the policy change. β_2 is the coefficient that directly measures the impact of the reform by estimating the differential trend observed in the treatment group during the post-implementation period ($D_o \times Post_t$). The regression further includes controls for the propensity score $pscore_{o,r}$ interacted with year dummies to control for any differential trends explained by differences in the propensity to enter the treatment group, occupation $\times$ region fixed effects ($\gamma_{o,r}$) to correct for time invariant market characteristics, and sector $\times$ year fixed effects ($\gamma_{s,t}$) to control

³⁶Appendix Table A9 shows that omitting to control for the propensity score creates some differences in pre-reform trends, which motivates the inclusion of this covariate in all the regressions of the paper. Similar differences arise also with the alternative treatment definition (table available upon request).

Table 1: Balancing test within establishments

	Estab - occup regressions				
	Constant		Treatment dummy		N. obs
Panel A: Stock variables					
Average levels					
Native empl.	17.97***	(1.095)	2.419**	(1.061)	407,708
Wage natives	29.68***	(0.679)	0.901	(0.656)	395,498
Residualized wage natives	0.382***	(0.0214)	0.0561***	(0.0190)	395,651
Migrant empl.	2.343***	(0.235)	-0.174	(0.204)	407,760
Wage migrants	24.87***	(0.818)	2.216**	(0.983)	82,187
Residualized wage migrants	0.291***	(0.0286)	0.110***	(0.0293)	82,232
Average yearly changes					
Native empl.	-0.664***	(0.0802)	-0.143	(0.0970)	303,181
Wage natives	0.490***	(0.0402)	0.0347	(0.0350)	292,751
Residualized wage natives	0.00189**	(0.000868)	0.000361	(0.000783)	293,356
Migrant empl.	-0.147***	(0.0223)	-0.00989	(0.0207)	303,357
Wage migrants	0.630***	(0.115)	0.0386	(0.124)	52,760
Residualized wage migrants	0.00889***	(0.00273)	-0.00283	(0.00296)	52,866
Panel B: Flow variables					
Average levels					
N. native hires	2.303***	(0.233)	0.301*	(0.183)	237,602
P. hiring a native	0.538***	(0.00563)	0.0301***	(0.00556)	237,602
N. migrant hires	0.177***	(0.0284)	0.00440	(0.0240)	237,602
P. hiring a migrant	0.0672***	(0.00493)	-0.00132	(0.00432)	237,602
Average yearly changes					
N. native hires	-0.533***	(0.114)	-0.0589	(0.0790)	77,909
P. hiring a native	-0.0314***	(0.00579)	0.00102	(0.00561)	78,002
N. migrant hires	-0.0628***	(0.0121)	0.0147	(0.0105)	77,903
P. hiring a migrant	0.00775***	(0.00220)	0.00220	(0.00190)	78,002

Each line corresponds to a separate regression where the outcomes are regressed on the treatment dummy over the period preceding the reform (2004-07), controlling for the propensity score and establishment fixed effects. Wages of incumbents are not reported because identical to the total wages in this period (the sample restriction only applies after the reform). In addition to the level of outcomes, I also report the effect on the yearly change. standard errors clustered at the region x occupation level. The constant term can be interpreted as the control average conditional on the fixed effects and the propensity score.

for any potential differences in exposure to the financial crisis driven by sector concentration. In the main specification, the identification comes from variation in how the outcome evolves over time between treatment and control occupations that are found within the same sector of activity, net of time invariant differences in region $\times$ occupation characteristics and of time trends explained

by each occupation's propensity to be part of the treated group. I further compare results obtained with and without the inclusion of establishment fixed effects (γ_i) to start getting a sense of the role played by firms in shaping the effect of the shock on wages.³⁷ The standard errors are clustered at the occupation $\times$ region level, and the time period considered goes from 2004 to 2013. Given the inclusion of multidimensional fixed effects, the analysis is performed using a linear estimator.

Given that the control group is also partially affected by an increase of supply driven by Bulgarian and Romanian workers, the estimates have to be interpreted as the effect of being exposed to a larger influx of immigrant workers. To obtain more economically meaningful magnitudes, I use the natural experiment to recover the elasticity of wages to the supply shock in the following 2SLS estimation:

$$L_{o,r,t}^M = \beta_0 + \beta_1 D_o + \beta_2 D_o \times Post_t + \sum_{t=1}^T \delta_t Pscore_{o,r} \times \gamma_t + \gamma_{o,r} + \gamma_{s,t} + \gamma_i + \epsilon_{o,i,r,s,t} \quad (2)$$

$$Y_{o,i,r,s,t} = \alpha_0 + \alpha_1 D_o + \alpha_2 \widehat{L}_{o,r,t}^M + \sum_{t=1}^T \phi_t Pscore_{o,r} \times \omega_t + \omega_{o,r} + \omega_{s,t} + \omega_i + \nu_{o,i,r,s,t} \quad (3)$$

Where $L_{o,r,t}^M$ captures the log of the stock of migrant employment within a given occupation and region, and is instrumented by $D_o \times Post_t$. α_2 recovers the elasticities of interest thus revealing the slope of the labor demand curve. I decide to aggregate migrant employment at the occupation $\times$ region level in order to avoid having a large number of zeros and to be able to directly recover elasticities. This aggregation entails little loss of information since the excluded instrument is itself aggregated. Nonetheless, I also show that results are robust to estimating the first stage on the log of total employment, including migrants and natives, at the establishment $\times$ occupation level ($L_{o,i,r,s,t}$).

³⁷This difference is suggestive of the importance of firms in shaping wages, but may also be due to different sources of identification exploited with and without fixed effects. The role of firms is thus further explored in the channel section.

5 Main Results

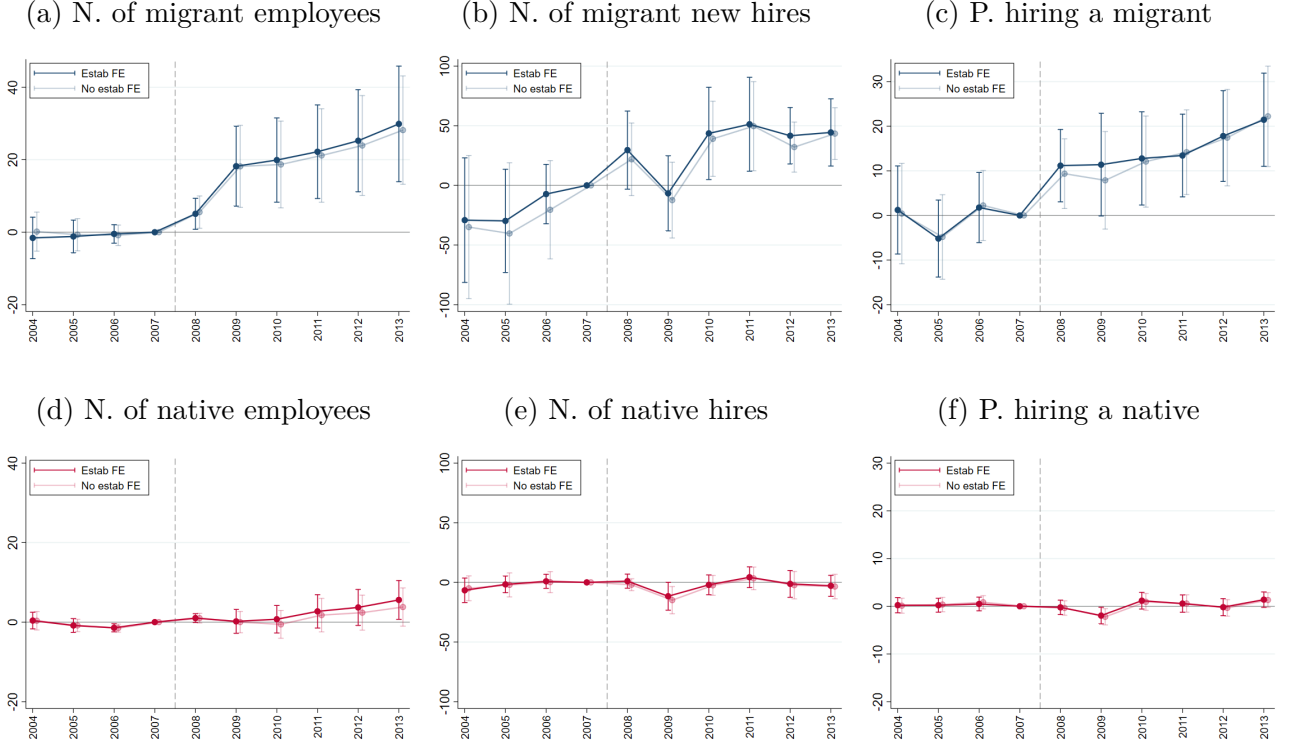
5.1 First stage: Employment

Figure 2 shows the dynamic graphs for the employment outcomes obtained from the model with and without establishment fixed effects. It plots the coefficients and the 95% confidence intervals obtained from a flexible model interacting each year dummy with the treatment group indicator, showing the estimated difference in trends between treatment and control occupations in each year of the sample with respect to the last year preceding the reform. To ease interpretation, the coefficients are rescaled to reflect percentage changes relative to the baseline mean. Treated and control groups follow similar trends before 2008 across all outcomes. After the reform, treatment occupations hire significantly more migrants than the control ones, and increase the stock of employment among migrants. Native employment does not change in the first years following the new policy, but it starts picking up from 2011 onward and becomes positive and significant in 2013, even though the effect remains much smaller than for migrants. This is consistent with a context where firms expand thanks to the reduction of their skill constraints (Clemens and Lewis, 2022). The effect on migrant employment appears to be long-lasting and there is no clear difference in the model with or without establishment fixed effects. Appendix Figure A7 shows that the results obtained using the alternative control definition are very similar.

Table 2 presents the main results obtained on hires and employment outcomes. Panel A shows the results obtained when controlling for establishment fixed effects and Panel B shows the results obtained when excluding establishment fixed effects. The full set of other fixed effects and controls described in Equation 1 is included in every regression. All outcomes are regressed in levels, since the large number of zeros present in the migrant variables preclude the use of logarithmic transformations. To interpret the magnitude of the coefficients, the table also reports the baseline means and calculates the effect in terms of growth relative to baseline (reported in bold when significant).

Results show that migrant employment reacts strongly and significantly to the policy change.

Figure 2: Dynamic graphs on employment growth relative to pre-reform mean



Note: The figure reports the estimated difference in trends between treatment and control occupations with respect to the last pre-reform year. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change relative to pre-reform mean, computed as the mean outcome observed within treated occupations before 2008. The light-colored line shows the results obtained without establishment fixed effects, while the dark-colored one shows the results obtained with establishment fixed effects.

Migrant employment stock grows by 21% on average during the six years following the reform relative to baseline, which is driven by a 51% increase in overall hires and a 16% increase in the pure extensive margin (probability of hiring). Omitting establishment fixed effects does not change the magnitude of the effects. Native employment does not change over the same period. Appendix Table A10 shows that the results obtained when using the alternative control definition are highly comparable.

Finally, Table A11 explores what happens if we look at the effect on the flows of EU migrants and on the flows of all migrants, EU and non-EU. As expected, the effect is much larger for non-EU born migrants: The N. of hires increases by 0.056 (+45%) among all migrants, driven by a 51% increase among non-EU migrants and a 26% among EU migrants. Nonetheless, the latter is significant. Similar differences appear on the probability of hiring a migrant. I can thus conclude

Table 2: Main results on employment

	(1)	(2)	(3)	(4)	(5)	(6)
	N. hires		P. hiring		Employment	
	Migrants	Natives	Migrants	Natives	Migrants	Natives
Panel A : with estab. fixed effects						
treat x post reform	0.0483** (0.0222)	-0.00412 (0.0588)	0.00693*** (0.00169)	0.000377 (0.00298)	0.262*** (0.0818)	0.454 (0.306)
Baseline trt mean	0.0938	2.089	0.0436	0.582	1.252	16.12
Effect in % growth	51%	0%	16%	0%	21%	3%
Observations	1,162,063	1,162,063	1,162,063	1,162,063	3,386,806	3,386,286
R-squared	0.214	0.289	0.195	0.202	0.249	0.246
Panel B : without estab. fixed effects						
treat x post reform	0.0523** (0.0259)	-0.0309 (0.0476)	0.00670*** (0.00202)	-0.00120 (0.00297)	0.249*** (0.0814)	0.313 (0.306)
Baseline trt mean	0.0938	2.089	0.0436	0.582	1.252	16.12
Effect in % growth	56%	-1%	15%	0%	20%	2%
Observations	1,194,530	1,194,530	1,194,530	1,194,530	3,396,974	3,396,455
R-squared	0.032	0.077	0.066	0.042	0.056	0.061

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

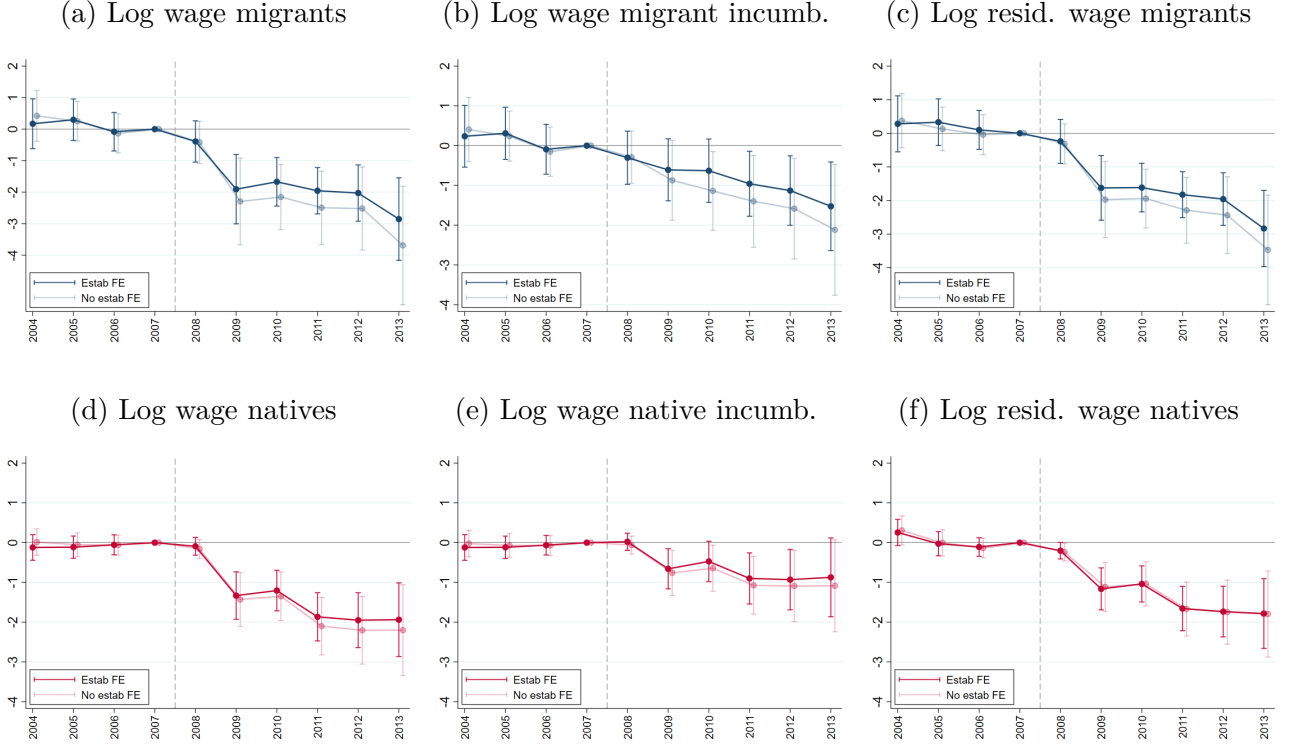
Standard errors clustered at the region-occupation level. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. Migrants refers to individuals born outside of France for the stock variables and to individuals born outside the EU for the flow variables. Natives refers to individuals born in France. Baseline mean reports the mean value of the outcome for treated occupations prior to the reform, and serves to compute the effect in terms of growth rate relative to baseline.

that the observed growth in migrant employment is by a large extent, even if not exclusively, explained by additional non-European hires.

5.2 Second stage: Wages

Figure 3 tests that pre-trends on the wage outcomes are parallel and shows how the effect evolved in the years following the reform both for the model with and without establishment fixed effects. Here as well there was no significant difference between treated and controls before 2008, while it becomes significantly negative afterwards. The negative impact on wages becomes smaller when I restrict the sample to individuals already working in France before 2008, but it remains significant

Figure 3: Dynamic graphs on wage growth



Note: The figure reports the estimated difference in trends between treatment and control occupations with respect to the last pre-reform year. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change, since coefficients are multiplied by 100. The light-colored line shows the results obtained without establishment fixed effects, while the dark-colored one shows the results obtained with establishment fixed effects.

and long-lasting. Finally, adding establishment fixed effects attenuates the effect, and particularly so for migrant wages. This finding suggests that establishments play a role in mediating the effect of supply shocks. Appendix Figure A8 shows the results obtained using the alternative control definition. The magnitudes of the negative wage effects are slightly smaller but broadly consistent with the baseline ones.

Table 3 shows the results of the regressions. All outcomes are inserted in a logarithmic form so that the coefficients can be interpreted as semi-elasticities. The table also computes the difference in the effect between migrants and natives and reports its significance level. When considering the specification with establishment fixed effects, I see that migrant salaries in reform occupations decrease by 1.8% after the shock, while native salaries only drop by 1.3%. The negative impact on wages is smaller when I restrict the sample to individuals already working in France prior to

the reform, reaching -0.9% for migrants and -0.5% for natives. The difference in the wage effect between the two types remain however strongly significant. The fact that the impact becomes smaller once we restrict to the incumbent sample can be due to two factors : i) the latter excludes composition changes driven by the supply shock, and ii) it also excludes the impact measured on the new graduates just entering the labor force. The residualized wage variable aims at controlling as much as possible for composition without excluding certain groups from the analysis and gives rise to coefficients in-between the first two. Interestingly, as already observed in Figure 3, we see that adding establishment fixed effects attenuates the negative effect on wages, and more so for migrants than for natives. Overall, regardless of the outcome considered, in the regressions with establishment fixed effects the impact on native wages is about two thirds of the one on migrant wages, while it gets closer to one half when establishment fixed effects are excluded. Appendix Table A12 shows that the wage results are similar when using the alternative control definition.

The finding that salary decreases in response to a positive shock in labor supply is in line with the predictions of the canonical model. The most striking fact about these results is the difference in the effect on migrant and native wages within affected occupations. This goes against the standard assumption of homogeneity between these two labor inputs when they are employed in the same occupations and in absence of occupation downgrading at migrants' arrival. Section 6 explores the channels that can explain such large differential effect. Finally, it is important to remember that here I am capturing the effect of the supply shock on the wages in affected occupations, to evaluate whether shortages might persist in the long run due to the deterioration of the native pay conditions. These results do not imply that affected workers, defined as the individuals that were in target occupations at the moment of the reform, suffer wage losses overall. In fact, if I assign treatment to individuals in 2006 according to their occupation and establishment at the time, and follow them up to 2010, assigning a wage of 0 to workers that exit the labor force, then I find a zero effect for both migrants and natives when I do not trim on establishment size in 2006, and a positive effect on native wages when I exclude establishments with less than 20 employees, as in the main analysis (see Appendix Figure A9 and Table A13). This result suggests that the negative wage effects found in the affected occupations do not translate into worse outcomes for

Table 3: Main results on wages

	(1)	(2)	(3)	(4)	(5)	(6)
	log wage		log wage incumbents		log wage residualized	
	Migrants	Natives	Migrants	Natives	Migrants	Natives
Panel A : with estab. fixed effects						
treat x post reform	-0.0183*** (0.00352)	-0.0133*** (0.00244)	-0.00900*** (0.00312)	-0.00550** (0.00263)	-0.0179*** (0.00294)	-0.0129*** (0.00241)
D Mig - Nat.	-0.00498*** (0.000174)		-0.00350*** (0.000169)		-0.00496*** (0.000147)	
Observations	592,182	3,264,838	526,473	3,020,920	592,436	3,266,321
R-squared	0.727	0.664	0.758	0.683	0.737	0.664
Panel B : without estab. fixed effects						
treat x post reform	-0.0240*** (0.00540)	-0.0158*** (0.00331)	-0.0131*** (0.00477)	-0.00743** (0.00326)	-0.0219*** (0.00466)	-0.0133*** (0.00334)
D Mig - Nat.	-0.00822*** (0.000468)		-0.00571*** (0.000350)		-0.00869*** (0.000360)	
Observations	605,562	3,275,266	538,684	3,031,020	605,812	3,276,739
R-squared	0.513	0.523	0.539	0.532	0.538	0.523

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. Migrants refers to individuals born outside of France, while natives refers to individuals born in France. Incumbents are individuals already working in France before 2008, while residualized wages control for age, gender and experience of workers within the establishment. D Mig - Nat computes the difference in the wage effect between migrants and natives, and report whether it is significant.

workers that were ex-ante exposed to the shock, once movers are included in the analysis.

5.3 Robustness Tests

Figure 4 reports the regression coefficients obtained from a battery of robustness tests on employment, wages and incumbent wages. Appendix Figure A10 shows the same tests for the N. of hires and the residualized wages. Each dot represents the result obtained from a separate regression, while the vertical line reports the 95% confidence interval. To ease interpretation, the employment coefficients are re-scaled by baseline means, such that all vertical axes can be read as growth rates. The coefficients in blue concern regressions with slightly modified samples or methodology, while

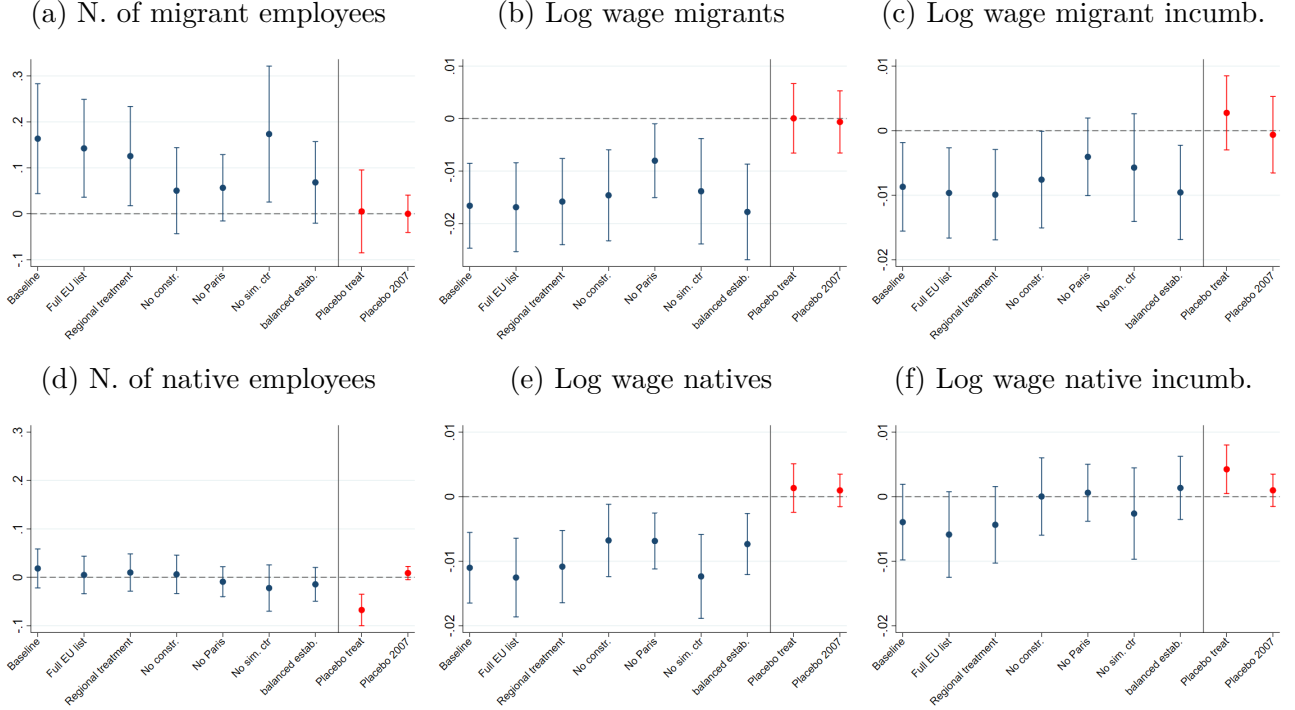
the ones in red are two distinct placebo tests.

The first column reports the baseline coefficient for comparison. The second column tests that the results are not driven by the sample restriction applied on control occupations in the main analysis. Here the coefficient is estimated using the full set of occupations in the EU list as controls, without excluding those with pre-reform tightness below the mean. The third column tests that results are robust to define treatment at the occupation $\times$ region level, thus excluding from the treatment group nationally treated occupations in regions where they do not appear in the local list. The fourth and fifth columns check that results hold when excluding *Ile de France*, the region where Paris is located, and the construction sector, which is directly targeted by some of the occupations of the list and is known for being an important employer of migrant labor, as well as the sector most affected during the financial crisis as shown by Appendix Figure A11. The sixth column evaluates the plausibility of the identification assumption regarding the absence of strong substitution effects in production between treated and control occupations. Here the coefficient is obtained after excluding from the control group the occupations that are the most likely to be substitutable to treated occupations (those that are the most similar as they share the first 3 digits out of the 4-digit classification with at least one treated occupation). Finally, the seventh column evaluates the role of establishment attrition by estimating the effect on establishments observed over the entire period.

The effect on employment of migrants becomes smaller and marginally insignificant when we exclude the region of Paris or the construction sector, which is in line with the fact that they concentrate an important share of migrant employment. The same is visible when looking at the number of migrant hires (marginally not significant only when construction is excluded). The effect on migrant and native wages also becomes smaller when Paris is excluded but remains significant, it becomes insignificant for incumbent wages but the effect on residualized wages remains stable across all tests. Excluding similar occupations from controls is expected to decrease the impact on employment and to increase the impact on wages in presence of substitution effects, since substitution would bias the results on employment upward and on wages downward.³⁸ Here,

³⁸The upward bias in employment comes from the fact that not only there is an increase in employment in treated

Figure 4: Robustness tests



Note: The figure reports the estimated effect of the policy across different specifications. The blue dots report the stability of the coefficients, while the red dots show the placebo tests. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change relative to pre-reform mean.

if anything, I observe the opposite, since the effect on migrant employment is slightly larger and the effect on migrant wages and on incumbent wages across both groups is slightly smaller. Finally, restricting the sample to a balanced panel of establishments shrinks slightly the employment effect but keeps most wage results unchanged. All in all, while some of the coefficients become smaller when we exclude parts of the sample, the overall message remains qualitatively similar across all these tests.

The last two columns of the figure, colored in red, perform two different placebo tests. The first placebo test only considers the observations in the control group and creates a placebo treatment indicator, taking the value of one for the 50% of the observations with the highest pre-reform levels of tightness. This is done to ensure that the main results are not driven by

occupations but also a decrease in employment in control ones, substituted away by treated, thus making the gap between the two larger. The downward bias on wages comes from the fact that wages in treated occupations are expected to decrease following the positive supply shock, but also the wages in control occupations would decrease following the decrease in demand generated by substitution, thus making the gap between the two smaller.

differences in preexisting skill shortages, rather than by the reform itself. Here all the coefficients are insignificant and close to zero in magnitude, except for the coefficient on native employment which is significantly negative. This reveals that, if anything, the higher level of tightness in the treatment group might introduce a downward bias on this outcome, which implies that the effect on native hires might be positive rather than insignificant. The second placebo test checks the effect detected for a placebo-reform set in the years preceding the actual policy change. In this analysis, 2007 is transformed to be a post-reform year and the model is estimated on the period going from 2004 to 2007. In this case all the coefficients are insignificant and close to zero, thus confirming the absence of anticipation effects.

Figures A12 and A13 in the Appendix evaluate the sensitivity of the employment and wage results to the inclusion of different levels of fixed effects. In the most demanding specification, I include establishment $\times$ year fixed effects, which absorbs both the sectoral and regional dimension, and only compares treated and control occupations that are found within the same plant. Results remain remarkably stable following the inclusion of increasingly demanding fixed effects. The fact that results on employment remain similar after controlling for establishment-specific shocks supports the hypothesis that they are not driven by the financial crisis, which is unlikely to have hit differently occupations within the same employer.

5.4 Magnitude of the Effect

The interpretation of the reduced form magnitudes is somewhat constrained by the fact that control occupations are also partly treated, given their exposure to Bulgarian and Romanian migrants. In order to obtain a more economically meaningful measure of the effect of the supply shock on wages, it is useful to compute the elasticity of wages to migrant employment, separately for migrants and natives. To obtain such measure I adopt the 2SLS procedure detailed in Equation 2, where the log of migrant employment in the region $\times$ occupation is predicted using the interaction between the treatment group and the post reform period, and is then regressed

Table 4: Wage elasticity

	(1)	(2)	(3)	(4)	(5)
	Log mig empl	Log wage Migrants	Natives	Log wage incumbent sample Migrants	Natives
	First stage	IV	IV	IV	IV
Panel A: with estab fixed effects					
treat x post reform	0.149*** (0.0228)				
Log mig empl		-0.147*** (0.0327)	-0.0912*** (0.0167)	-0.0755*** (0.0254)	-0.0388** (0.0170)
Observations	3,374,729	592,182	3,252,602	526,473	3,010,000
R-squared / RMSE	0.984	0.257	0.270	0.238	0.257
K-P F statistic		21.38	42.59	20.56	41.80
Panel B: without estab fixed effects					
treat x post reform	0.158*** (0.0254)				
Log mig empl		-0.168*** (0.0333)	-0.102*** (0.0181)	-0.0956*** (0.0290)	-0.0492*** (0.0188)
Observations	3,384,917	605,562	3,263,054	538,684	3,020,104
R-squared / RMSE	0.982	0.326	0.316	0.312	0.306
K-P F statistic		22.20	38.57	21.88	38.23

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. The table presents the IV results where the log of migrant employment in a given occupation x region is instrumented using the interaction between the treatment indicator and the post-reform period. The outcome wages are measured within occupations and establishments. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. The bottom of the page reports the Kleibergen-Paap F-statistic.

on wages at the establishment $\times$ occupation level.³⁹ All controls used in the main analysis are included here as well.

Results are reported in Table 4 and in Appendix Table A14 using alternative controls. The first stage is presented in Column (1): the reform increases migrant employment in affected occupations $\times$ regions by about 15%, which is broadly consistent with the results shown in Table

³⁹I aggregate the first stage at the region $\times$ occupation level to be able to take the logarithm, avoiding a large number of zeros. Since the excluded instrument is itself at an aggregate level, this simplification results in little loss of information for the second stage.

2. A 1% increase in the stock of migrant employment within technical occupations decreases migrant wages by 0.15% and native wages by 0.09%. The elasticity of incumbent wages is slightly smaller, 0.07% and 0.04% for migrants and natives respectively. As already suggested by the reduced form results, the elasticity of wages, and of migrant wages in particular, is larger when we do not account for establishment fixed effects. The same analysis using the alternative controls gives rise to qualitatively similar results.

Finally, as an additional robustness test keeping the first stage outcome at the level of the establishment $\times$ occupation, I compute the elasticity of wages to changes in the log of total employment, including migrants and natives. Results are reported in Appendix Table A15. The first stage shows that the reform raises employment in treated occupations by about 4%, which is another indication of the success of the policy for increasing the availability of these technical skills for firms. Secondly, the elasticity of wages relative to overall employment is as expected larger, given that the magnitude of the first stage is smaller : a 1% increase in employment leads to a 0.45% decrease in migrant wages and a 0.3% decrease in native wages. The elasticity for incumbent wages is about half.

6 Channels explaining the differential wage effect

Section 5 shows that the drop in migrant wages is more than 50% larger than the one in native wages following the migrant supply shock, which translates into a migrant wage elasticity more than 50% larger than the native one. This section tests three possible channels that can explain such difference: i) composition effects, ii) differential sorting across establishments, iii) differential effects within establishments.

6.1 Composition Effects

The first reason that could explain the differential wage effect observed between migrants and natives is a change in the composition of workers following the reform. The new migrants hired through the policy might be younger and less experienced than the migrants that were already present in the country. Young natives might escape the competition by switching to other occupations not accessible to migrants, such that the change in composition attenuates the negative wage effect among them (Foged and Peri, 2016).

The analysis on the wages of incumbents already present in the country before 2008 excludes any change in composition driven directly by newly arrived migrants, but might still be subject to composition changes driven by better paid workers moving to other occupations and establishments.⁴⁰ The results on residualized wages control for changes in composition according to the age, the gender, and the years of experience of workers. To further evaluate the extent of composition changes in the overall sample and in the incumbent sample, I use the same identification as in the main analysis to evaluate the impact of the reform on the average age, the share of women and the average experience within new hires and employment, separately for migrants and for natives. Second, I evaluate the presence of changes in mobility patterns by looking at the effect of the reform on the rate of employer switches and of employer and occupation switches.

Table A16 in the Appendix reports the results. Within the hiring flows, migrants and natives are slightly more likely to be women after the reform, while no change is observed on their average age. Within the employment stocks, migrants and natives become slightly younger, and natives also less experienced. While these differences are significant they are small in magnitude. Additionally, given that similar patterns arise in both the migrant and native samples, these changes are unlikely to explain the differential wage effect across them. Finally, the differences in characteristics become smaller for workers already present in France before 2008, confirming that looking at the wages of incumbents abstracts by a large extent from composition changes.

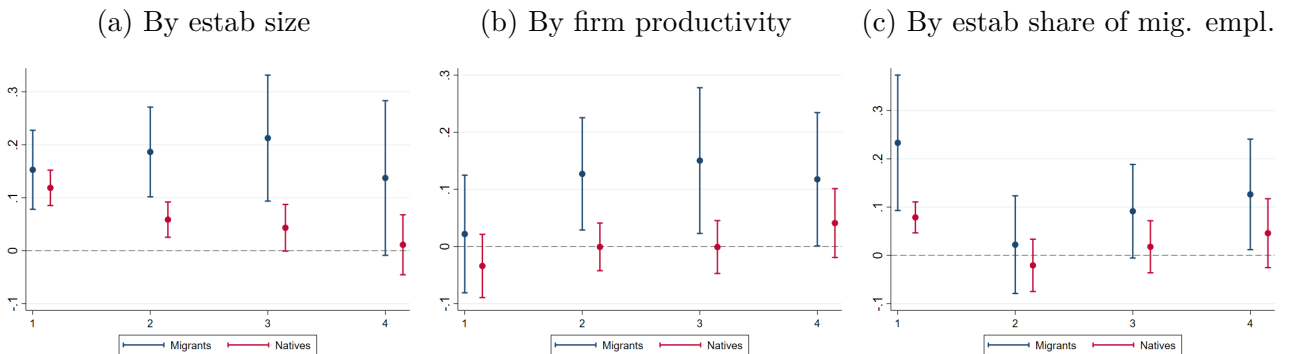
⁴⁰Note that given that I fixed the occupation based on the one of first appearance in the establishment, the only way to exit the treated sample is to switch both occupation and employer.

Table A17 shows the effects of the reform on the establishment switching rate and the establishment-occupation switching rate. Mobility rates increase by 3% among both migrants and natives in regressions with establishment fixed effects. This is consistent with the fact that lower tightness increases turnover. Nonetheless, once establishment fixed effects are omitted, the effect on all types of mobility goes to zero, confirming that such mechanism is unlikely to generate large changes in composition within the broad treatment group.

6.2 Differential Sorting across Establishments

A second mechanism that could explain the differential wage effect among natives and migrants concerns sorting across establishments and firms (Arellano-Bover and San, 2020). Recent work by Brinatti and Morales (2021) in Germany shows that migrants tend to sort into larger and more productive firms, while natives respond by relocating towards native-intensive firms that specialize in different goods and services. This endogenous reallocation reduces the degree of competition that migrants exert on natives, thus increasing the overall welfare gains of immigration. The fact that the negative wage effect obtained without establishment fixed effects is larger than the one obtained with these fixed effects suggests that differential sorting across firms might play an important role.

Figure 5: Employment effects by quartile of employer characteristics



Note: The figure reports the estimated effect of the policy across quartiles of establishment size, firm productivity, and establishment share of migrant employees in all other occupations. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change relative to pre-reform mean.

To test this mechanism further, I divide the sample in quartiles across three different establishment

characteristics, measured in the first year of establishment appearance in the data: i) size, ii) productivity,⁴¹ and iii) share of employment done by migrants in occupations outside the reform list. I then estimate the effect of the reform on migrant and native employment within each quartile separately, to see whether they differ.

Results reported in Figure 5 show that the effect on migrant employment is only present in firms above the first quartile of productivity, but is rather constant across the remaining three quartiles. The smallest and largest establishments show a somewhat smaller increase in migrant employment relative to medium-sized establishments. The increase in migrant employment is also larger in establishments with a low initial share of migrants measured in non-treated occupations, which suggests that the reform increased access to migrant employment for plants that were not previously able to pay the cost. Interestingly, I do find, as Brinatti and Morales (2021), that native employment increases in establishments of small size while it stays unchanged among larger ones. This suggests that part of the reduced form negative effect on native wages is explained by natives becoming relatively more concentrated in smaller establishments, which rationalizes the result that controlling for establishment fixed effects makes the wage impacts smaller. Appendix Table A18 re-estimates the main model within size, productivity and share of migrants quartiles by adding quartile fixed effects interacted with year dummies.⁴² The coefficients remain broadly similar to the ones obtained in the baseline analysis, but the gap in the wage effect between migrants and natives becomes larger. In fact, the effect on native incumbent wages becomes insignificant when the comparison with the control group is restricted to the same quartile of firm characteristics. It seems thus that most of the wage penalty suffered by natives is explained by their sorting into smaller plants. Controlling for this sorting effect exacerbates the disparity in the impact on natives and migrants, rather than explaining it.

⁴¹Firm productivity is computed as the firm value added divided by the total number of employees. Firm value added is obtained from an additional administrative data source containing firms' financial records and called *FICUS-FARE*. Given that this data does not distinguish between establishments of the same firm, I assign the same productivity to all of them.

⁴²The interaction with year fixed effects allows me to control not only for the effect of the levels but also of the trends.

6.3 Differential Effects within Establishments

The final channel that could explain the different wage effect between natives and migrants is related to mechanisms taking place within establishments. Consistent with a large body of empirical literature, it might be that migrants and natives are imperfect substitutes in production,⁴³ even in this somewhat extreme context where they are employed within the same occupations and by the same plants. Additionally, if we relax the assumption of perfect competition and allow firms to extract rents, it might be that natives have higher bargaining power with respect to migrants and are thus able to attenuate negative wage pressures. This is likely to be the case in France, given that if migrants wish to switch employer, they have to obtain a visa sponsored by the new employer, which limits considerably their outside options (Naidu, Nyarko and Wang, 2016).

Several papers have estimated the elasticity of substitution parameter using nested CES production functions.⁴⁴ However, these models assume perfect competition in the labor market. Amior and Manning (2020) revisit the model by allowing for monopsonistic competition. In their production function the output Y_t of a representative firm is produced using labor inputs from different skill groups indexed by o , which are assembled through a CES production function:

$$Y_t = A_t \left[\sum_o \theta_{ot} \mathbf{L}_{ot}^{\frac{\omega-1}{\omega}} \right]^{\frac{\omega}{\omega-1}} \quad (4)$$

where ω is the elasticity of substitution between the skill groups and θ_{ot} is the skill group-specific efficiency parameter that might vary through time. Further, the labor in each skill group L_{ot} combines inputs from native workers N_{ot} and migrant workers M_{ot} through a CES production function nested into the first level:

$$\mathbf{L}_{ot} = \left[\beta_{ot} M_{ot}^{\frac{\sigma-1}{\sigma}} + N_{ot}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (5)$$

⁴³See for instance Peri and Sparber (2009); D'Amuri, Ottaviano and Peri (2010); Lewis (2013).

⁴⁴Notably Card (2009); Manacorda, Manning and Wadsworth (2012); Ottaviano and Peri (2012).

where σ is the elasticity of substitution between natives and migrants, and β_{ot} is the relative efficiency of migrants with respect to natives within each skill group.⁴⁵

Differently from the perfect competition model, in [Amior and Manning \(2020\)](#) each firm faces a labor supply curve that is upward sloping in wages. Appendix C lays out the details of the firm optimization problem, which results in equilibrium wages that equal the marginal product of migrant and native labor, minus their specific mark-down (ϕ_{ot}^M and ϕ_{ot}^N). This allows them to express the relative wage of migrants to natives using the following formula:

$$\ln\left(\frac{w_{ot}^M}{w_{ot}^N}\right) = \ln(\beta_{ot}) - \Delta\phi_{ot,0} - \left(\frac{1}{\sigma} + \Delta\phi_{ot,1}\right)\ln\left(\frac{M_{ot}}{N_{ot}}\right) \quad (6)$$

Where $\Delta\phi_{ot,0}$ is the difference in mark-down between migrants and natives that does not depend on the migrant share, and $\Delta\phi_{ot,1}$ is the difference in mark-down between migrants and natives that does depend on the migrant share and thus might be impacted by the reform. This formula reveals that, once the assumption of equal mark-downs is relaxed, it is not anymore possible to disentangle the elasticity of substitution parameter (σ) from the differential mark-down parameter that depends on the migrant share ($\Delta\phi_{ot,1}$). Nevertheless, the empirical estimation of Equation 6 allows to recover the combined effect of these two factors, and therefore to rule out the presence of perfect competition and perfect substitutability combined.

To estimate Equation 6 econometrically, similarly to the estimation of the wage elasticities presented in Section 5.4, I take advantage of the reform as an exogenous instrument. Here the instrumented variable is the ratio of the supply of migrant to native employment. In addition, to avoid restricting the analysis on the establishment $\times$ occupation cells in which both migrants and natives are observed, I split the estimation of Equation 6 in two separate regressions⁴⁶ :

⁴⁵The existence of imperfect substitutability between migrants and natives within cells can be rationalized by the fact that each skill group is an aggregation of unobservable tasks in which the distribution might differ between natives and migrants due to differences in comparative advantage.

⁴⁶This approach is similar to the one adopted in [Acemoglu, Autor and Lyle \(2004\)](#).

$$\ln(L_{o,r,t}^{M,N}) = \beta_0 + \beta_1 D_o \times Post_t + f_i + f_{o,r} + f_{s,t} + v_{o,i,r,s,t} \quad (7)$$

$$\ln(w_{o,i,r,s,t}^M) = \alpha_0^M + \alpha_1^M \widehat{\ln(L_{o,r,t}^{M,N})} + f_i + f_{o,r} + f_{s,t} + \epsilon_{o,i,r,s,t} \quad (8)$$

$$\ln(w_{o,i,r,s,t}^N) = \alpha_0^N + \alpha_1^N \widehat{\ln(L_{o,r,t}^{M,N})} + f_i + f_{o,r} + f_{s,t} + \epsilon_{o,i,r,s,t} \quad (9)$$

where $L_{o,r,t}^{M,N}$ is the employment ratio of migrants to natives within an occupation o , region r and time t , instrumented by the interaction between the treatment status and the post-reform period ($D_o \times Post_t$). $\ln(w_{o,i,r,s,t}^M)$ and $\ln(w_{o,i,r,s,t}^N)$ are the log of migrant and native wages in establishment i , occupation o , sector s and time t . The parameter of interest $-(\frac{1}{\sigma} + \Delta\phi_{ot,1})$ is obtained by computing $\alpha_1^M - \alpha_1^N$ from Equations 8 and 9. Intuitively, I estimate the effect that a change in the ratio of migrants to natives has on the wage of migrants and natives, and I see whether it differs.

Column (1) of Table 5 shows the first stage results. Columns (2) and (3) estimate the effect on migrant and native wages and columns (4) and (5) look at the wages of incumbents. Appendix Table A19 shows the OLS results and Appendix Table A20 computes the same analysis restricting to establishments that hire both natives and migrants. The reform has a positive and significant effect on the log ratio of migrant to native employment in the occupation and region (+3.3%). The IV results show that a 1% increase in the ratio of migrant to native employment decreases migrant wages by 0.47% and native wages by 0.37%. It also decreases incumbent migrant wages by 0.26% while the effect on incumbent native wages is not significant. Both of these results are suggestive that the combined assumption of perfect substitutability and similar bargaining power can be rejected, even if two caveats have to be noted: The Kleibergen-Paap F-statistic of the first stage is below the conventional levels to consider the instrument strong, which also affects the precision of the second stage coefficients such that the difference between them is not significant at conventional levels. I thus take these results as suggestive. The effects are similar if I restrict the sample to establishments hiring both natives and migrants (Table A20), while the OLS coefficients are much smaller in magnitude but confirm the larger negative effect on migrant

wages (and here the difference is significant at the 1% level).

Table 5: Elasticity of substitution between migrants and natives

VARIABLES	(1)	(2) (3)		(4)	(5)
	Log ratio mig to native empl.	Log wage		Log incumbent wage	
		Migrants	Natives	Migrants	Natives
	First stage	IV	IV	IV	IV
treat x post reform	0.0333** (0.0164)				
Log ratio migrant to native empl.		-0.471** (0.211)	-0.370** (0.175)	-0.263** (0.130)	-0.131 (0.0927)
D Mig - Nat. $-(\frac{1}{\sigma} + \Delta\phi_{ot,1})$		-0.102 (0.700)		-0.133 (0.272)	
Observations	3,369,478	589,021	3,247,656	523,685	3,005,526
R2 / RMSE	0.946	0.264	0.279	0.240	0.258
K-P F-statistic		4.028	3.898	3.484	3.801

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. The table presents the IV results where the log ratio of migrant employment over native employment in a given occupation x region is instrumented using the interaction between the treatment indicator and the post-reform period. The outcome wages are measured within occupations and establishments. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Column (1) shows the first stage regression. Columns (2) and (3) present the IV estimation on migrant and native wages, and columns (4) and (5) present the IV estimation on incumbent migrant and native wages. The bottom of the page reports the Kleibergen-Paap F-statistic for the instrument relevance. The table also reports the difference between the migrant and native wage effects and the implied parameter capturing the elasticity of substitution + the difference in bargaining power between migrants and natives.

Finally, to get a sense of whether bargaining power plays a role beyond the elasticity of substitution factor, I test how the effects vary in contexts where establishments are more or less likely to enjoy some monopsony power over natives. In the first case, natives are expected to have outside options that are as limited as those of migrants and might therefore suffer in a more similar manner from the increase in competition ($\Delta\phi_{ot,1}$ is small). The opposite happens in contexts with low monopsony power. Differences in monopsony power are not expected to impact the technological substitutability σ , thus identifying the effect of changes in $\Delta\phi_{ot,1}$ alone.

The first split of the data divides establishments across the share of total employment that they hire within a given occupation and region, computed before the policy shock. Establishments with higher shares are expected to have larger monopsony power. The second split is done according to the level of employers' concentration across occupations and regions, computed based on an Herfindahl Index. The third split is done according to the rate of job to job mobility in a given region and occupation, with the idea that the easier it is to move to a different employer, the lower the exposure of native workers to rent extraction from employers. Appendix Table A21 shows that

the difference in the effect on migrant and native wages is larger within employers concentrating a lower share of occupation-region employment and in occupations and regions with lower overall employment concentration. This is not true for contexts with high job-to-job mobility but here the effect on native wages is not significant. The instrument strength remains an issue even when selecting the sub-sample with the largest first stage power, so these results have to be taken as suggestive that the bargaining power channel could explain part of the differential effects shown in the main analysis.

7 Conclusion

This paper shows that relaxing constraints on immigrant visas targeted to occupations facing skill shortages is an effective tool to increase labor supply in these jobs and does not harm native employment prospects. Additionally, native wages in reform occupations are partly shielded from the additional competition, thus suggesting that the reduction of labor market shortages have the potential to last in the long term. Further analysis should be undertaken to test whether, in the long run, these kinds of policies can discourage countries' adaptation of their national education systems. An important criticism of this policy resides in the fact that the list remained unchanged up to 2021, while the needs of the labor market have significantly evolved over the past ten years. The periodic adaptation of the list of occupations facing skill shortages, as it is done in the UK for instance, might help maximize the support given to domestic firms and minimize the cost for domestic labor.

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Online Appendix for

Do Skilled Migrants Compete with Native Workers?

Analysis of a Selective Immigration Policy

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Outline

- Appendix [A](#) presents additional tables and figures
- Appendix [B](#) presents the data in more details
- Appendix [C](#) presents the details of the model introduced in section 6.3

Appendix

A Additional Tables and Figures

A.1 Figures

Figure A1: Administrative Procedure to Hire a Foreign Worker

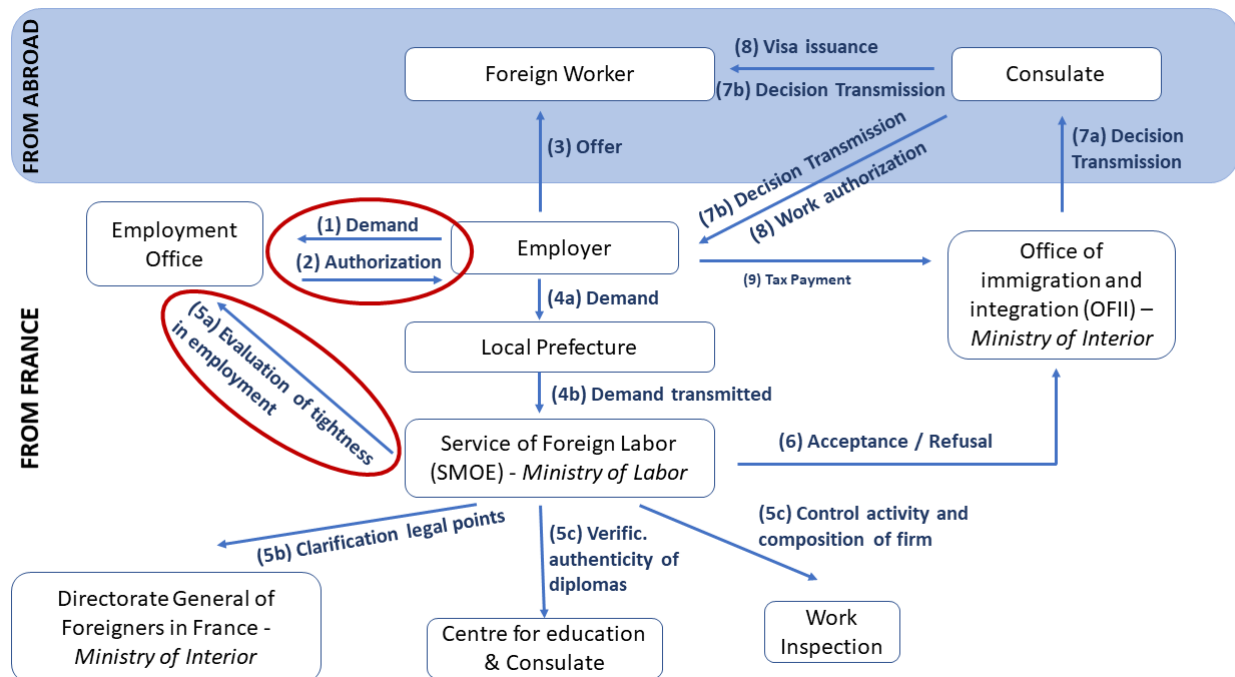
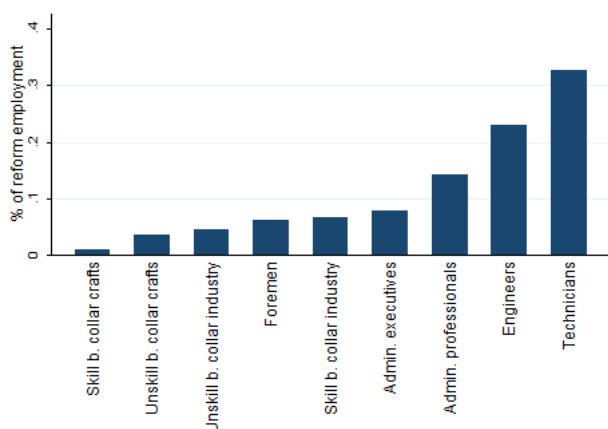


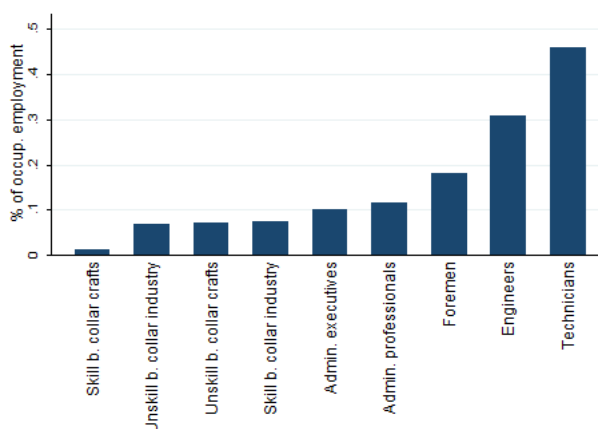
Diagram taken from [OECD \(2017\)](#). The red circles identify the administrative steps relaxed by the reform.

Figure A2: Occupational characterization of the reform

a) Occup. distribution of reform employment



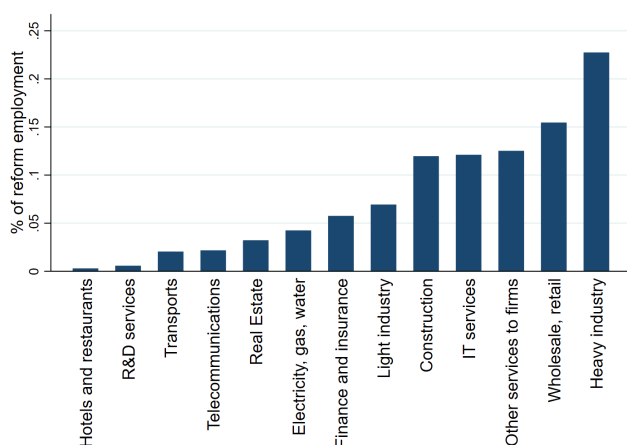
b) Occupational exposure to the reform



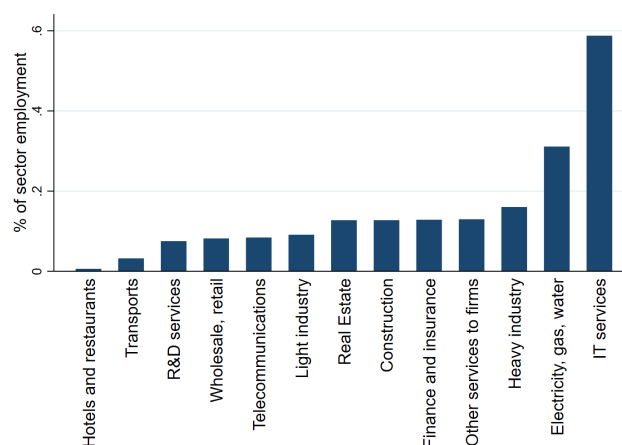
Author's calculations based on data from French Labour Force Survey 2004-2007. Panel a) shows how the total employment concerned by the reform is distributed across broad occupation categories. Panel b) shows how much of each occupation's employment is concerned by the reform (measure of exposure).

Figure A3: Sectoral characterization of the reform

a) Sectoral distribution of reform employment



b) Sectoral exposure to the reform



Author's calculations based on data from French Labour Force Survey 2004-2007. Panel a) shows how the total employment concerned by the reform is distributed across sectors. Panel b) shows how much of each sector's employment is concerned by the reform (measure of exposure).

Figure A4: Immigration flows by year and broad category

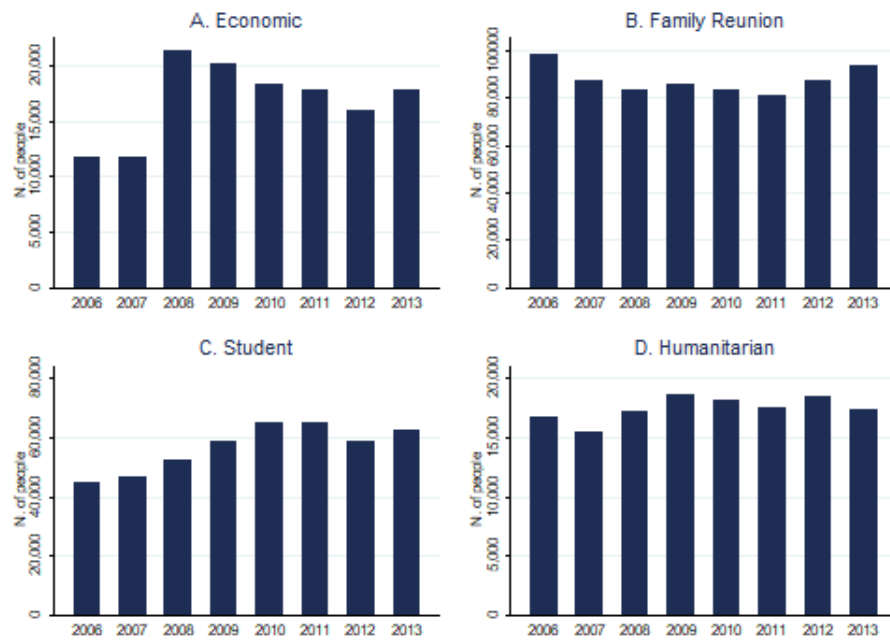
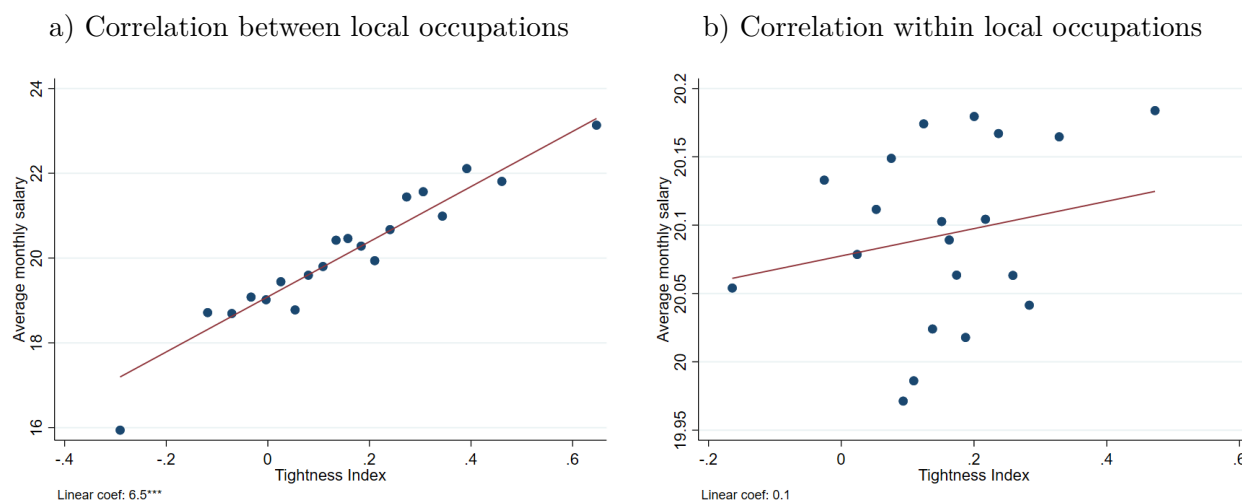
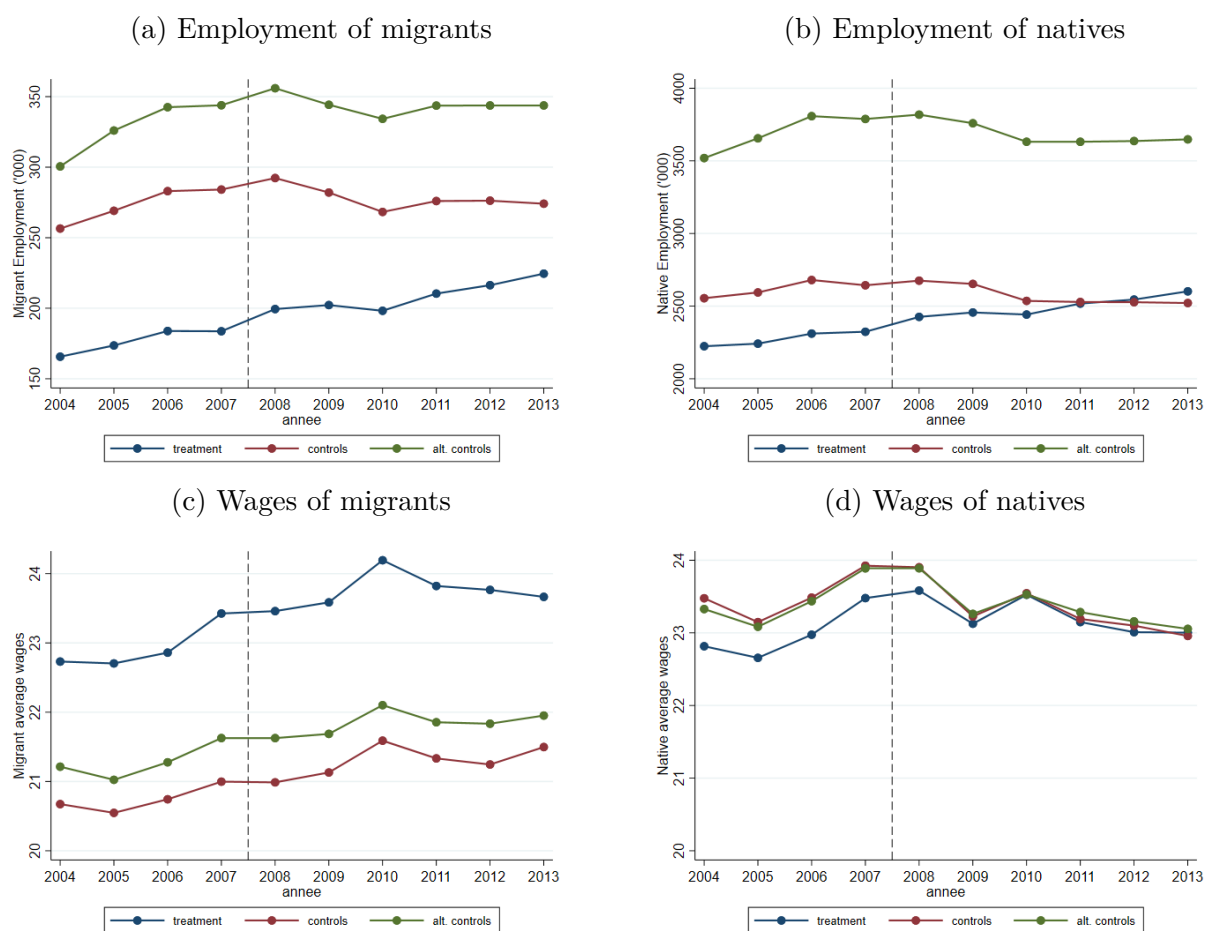


Figure A5: Correlation between local occupation wage and tightness



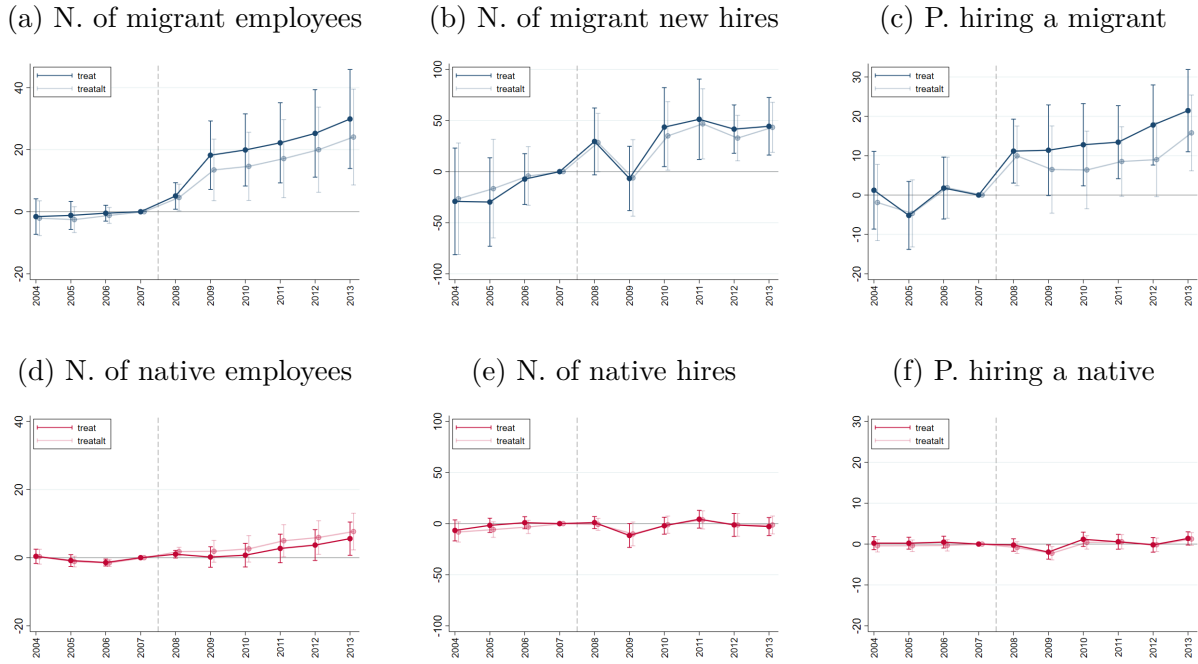
Each dot bins together similar observations at the level of occupation $\times$ region. Average wages within occupations and regions are computed using payroll tax data (DADS poste), and occupation $\times$ region level of tightness is taken from data produced by the French Employment Office. The graph shows the binned scatter plot and reports the coefficient obtained from the linear regression. Panel A shows the correlation between wage and tightness using cross-sectional variation across occupations and regions. Panel B controls for occupation $\times$ region fixed effects, and thus exploits variation within them across time. The period used goes from 2004 to 2007 and the occupations are restricted to those targeted by the reform.

Figure A6: Aggregate trends in outcomes



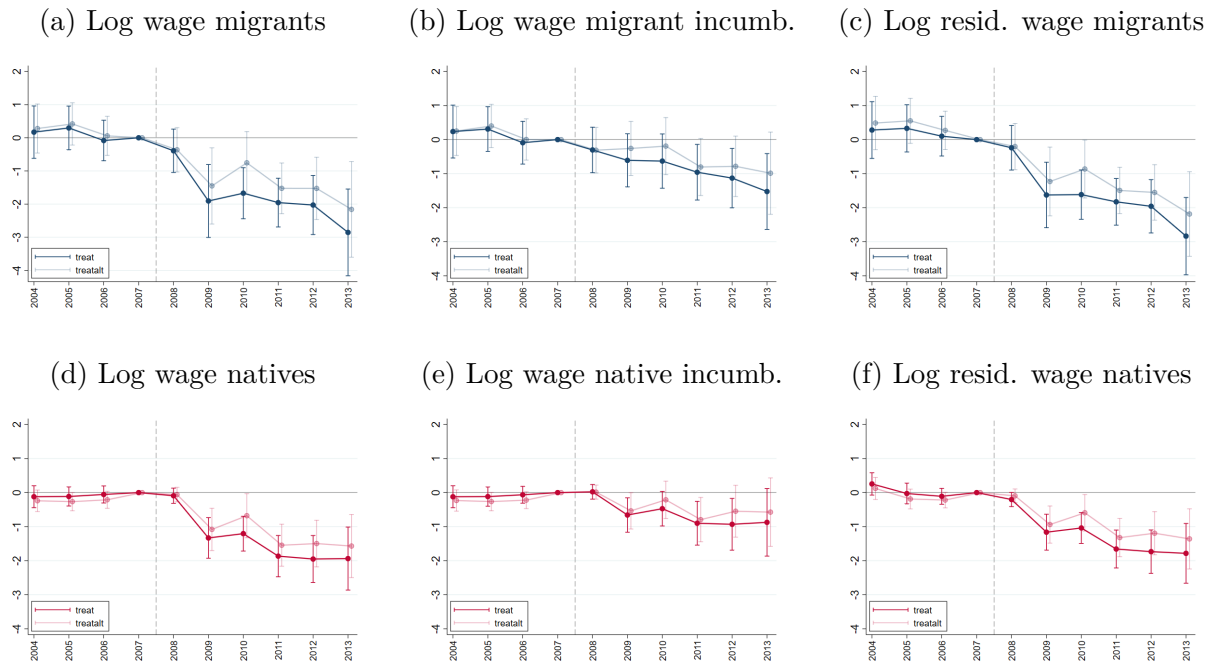
The figure shows the total employment of migrants and natives expressed in thousands of full time equivalent and the average wage of migrants and natives. The three categories express the treatment group, the main control group (list of occupations open to Romanians and Bulgarians), and the alternative control group (all occupations in the top two quantiles of the tightness distribution).

Figure A7: Dynamic graphs on employment growth relative to pre-reform mean using alternative controls



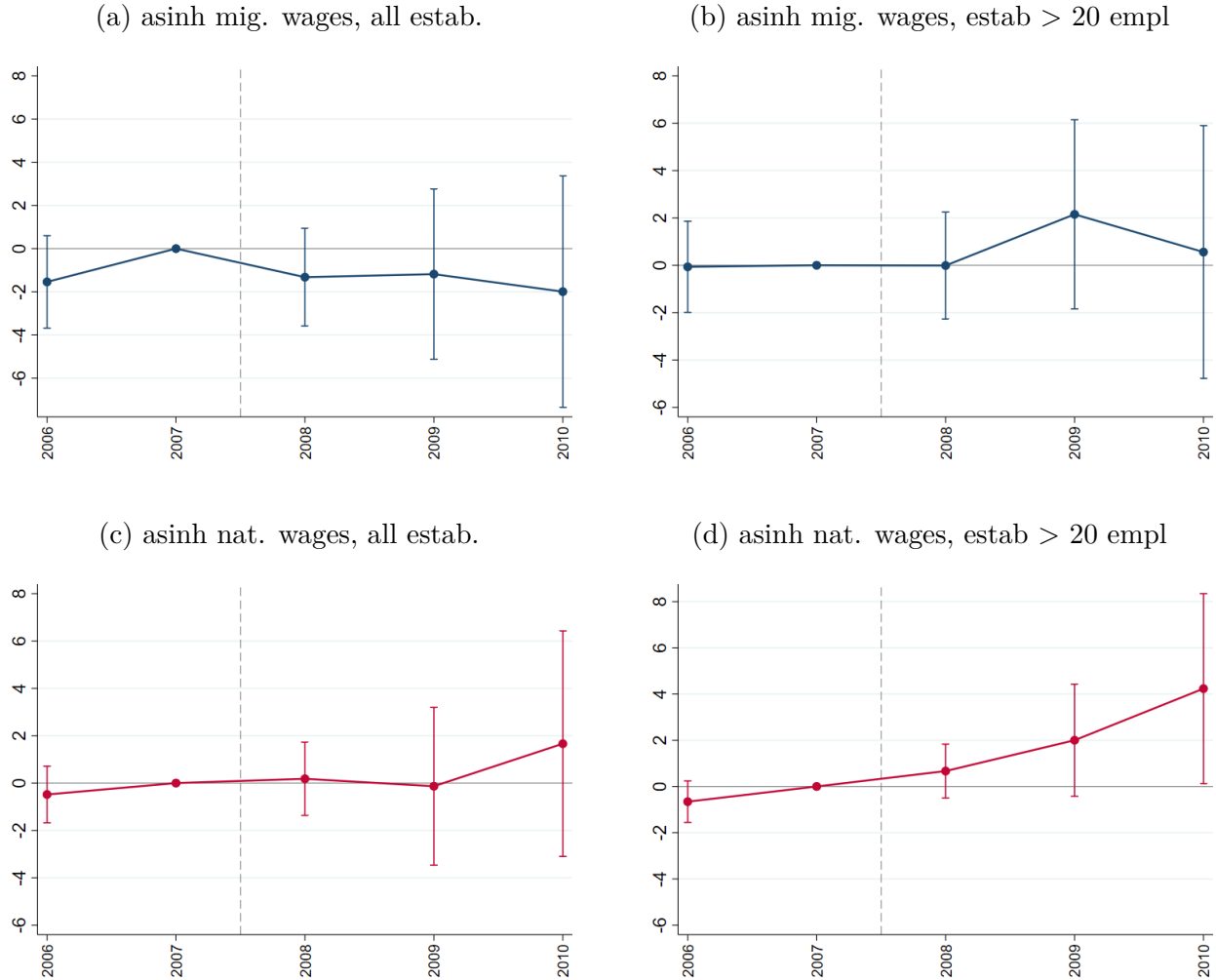
Note: The figure reports the estimated difference in trends between treatment and control occupations with respect to the last pre-reform year. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change relative to pre-reform mean, computed as the mean outcome observed within treated occupations before 2008. The light-colored line shows the results obtained using the alternative control group, while the dark-colored one shows the results obtained with the baseline treatment definition. Both models include establishment fixed effects.

Figure A8: Dynamic graphs on wage growth using alternative controls



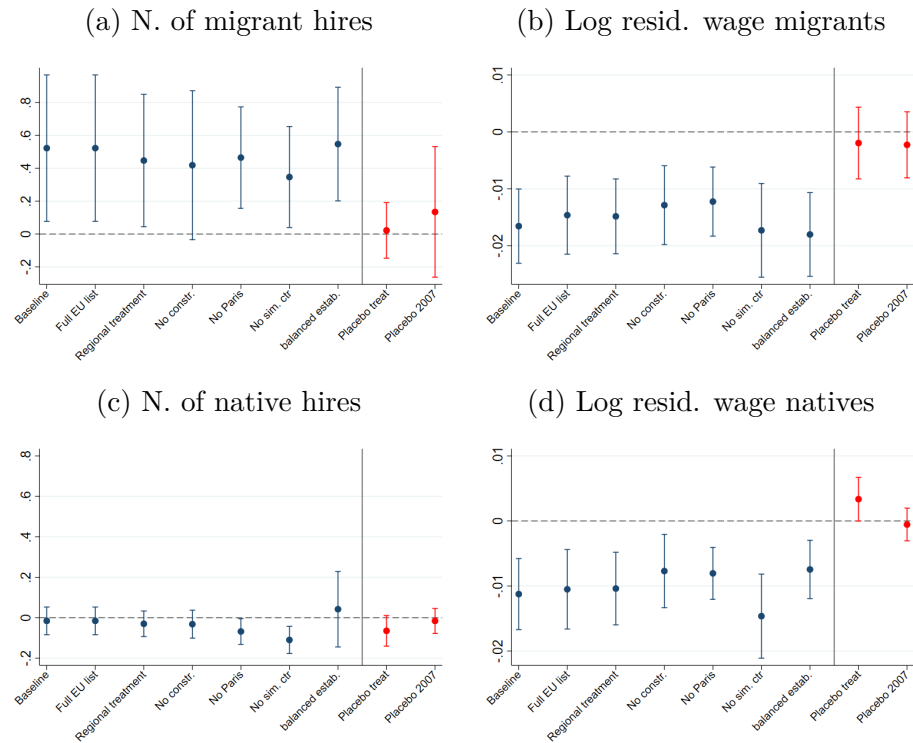
Note: The figure reports the estimated difference in trends between treatment and control occupations with respect to the last pre-reform year. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change, since coefficients are multiplied by 100. The light-colored line shows the results obtained using the alternative controls, while the dark-colored one shows the results obtained with the baseline treatment definition. Both models include establishment fixed effects.

Figure A9: Dynamic graphs on wages for workers in treatment and control occupations in 2006 followed over the entire period



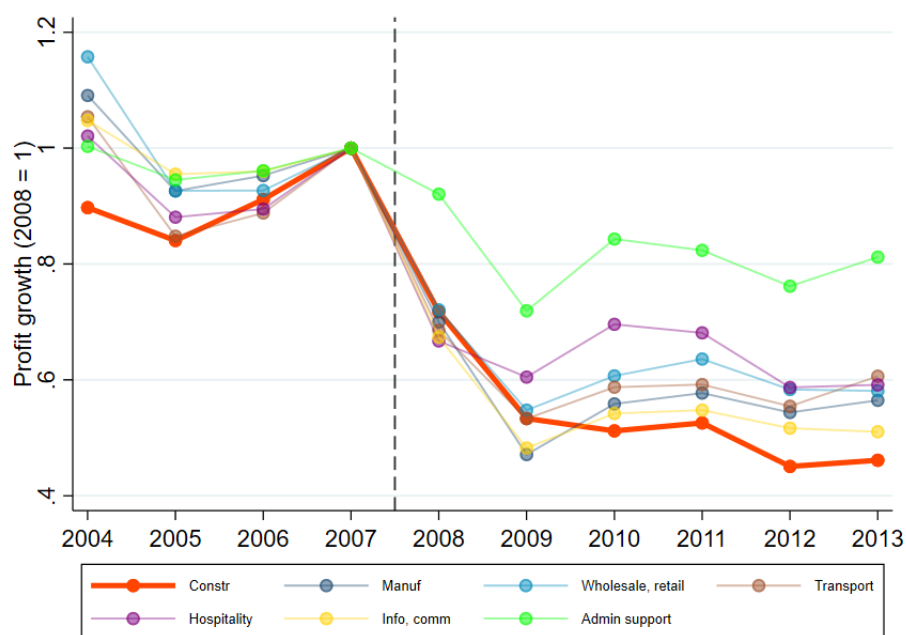
Note: The figure reports the estimated difference in trends between treatment and control workers with respect to the last pre-reform year. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change, since coefficients are multiplied by 100. The sample is restricted to individuals that are observed in a treated or control occupation in 2006 and that are followed over the subsequent years, even if they change occupation, establishment or they exit the labor force (in which case I assign them a wage of zero). I then re-aggregate observations at the occupation x establishment cells, based on those observed in 2006. The outcomes are transformed using the hyperbolic sine to keep the 0s in the analysis. I present both the results dropping workers that in 2006 were employed in firms with less than 20 employees, to follow closely my main analysis, and by keeping them in.

Figure A10: Robustness tests on additional outcomes



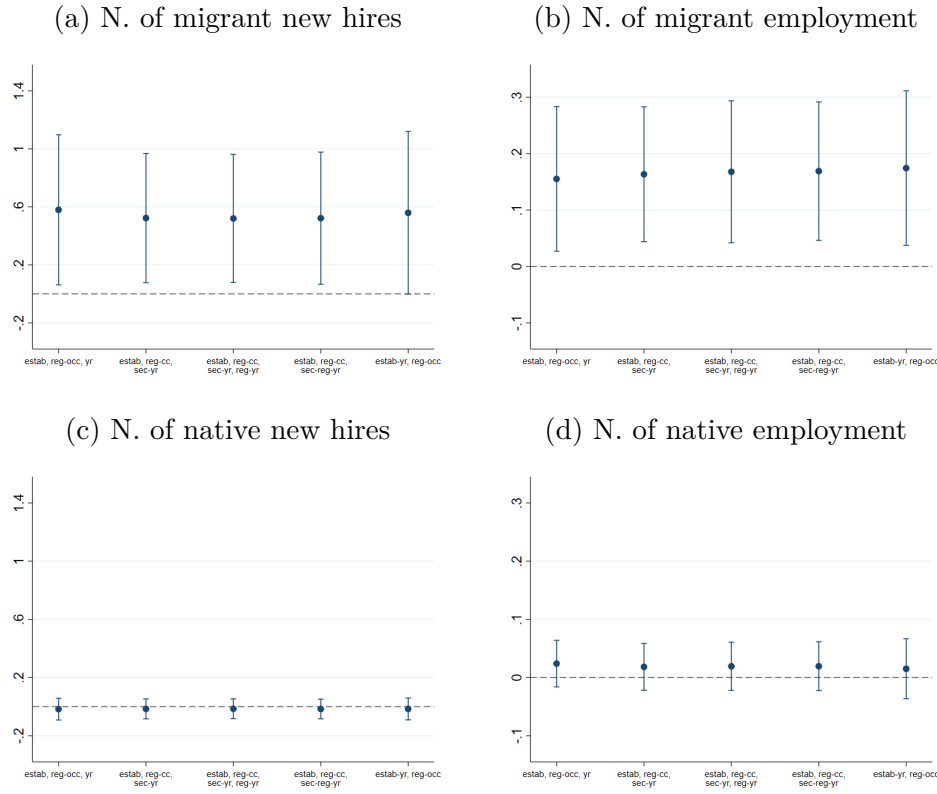
Note: The figure reports the estimated effect of the policy across different specifications. The blue dots report the stability of the coefficients, while the red dots show the placebo tests. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change, since coefficients for level outcomes are divided by the pre-reform mean, computed as the mean outcome observed within treated occupations before 2008, and all coefficients are multiplied by 100.

Figure A11: Growth in profits by broad sector, where 2008 is normalized to 1



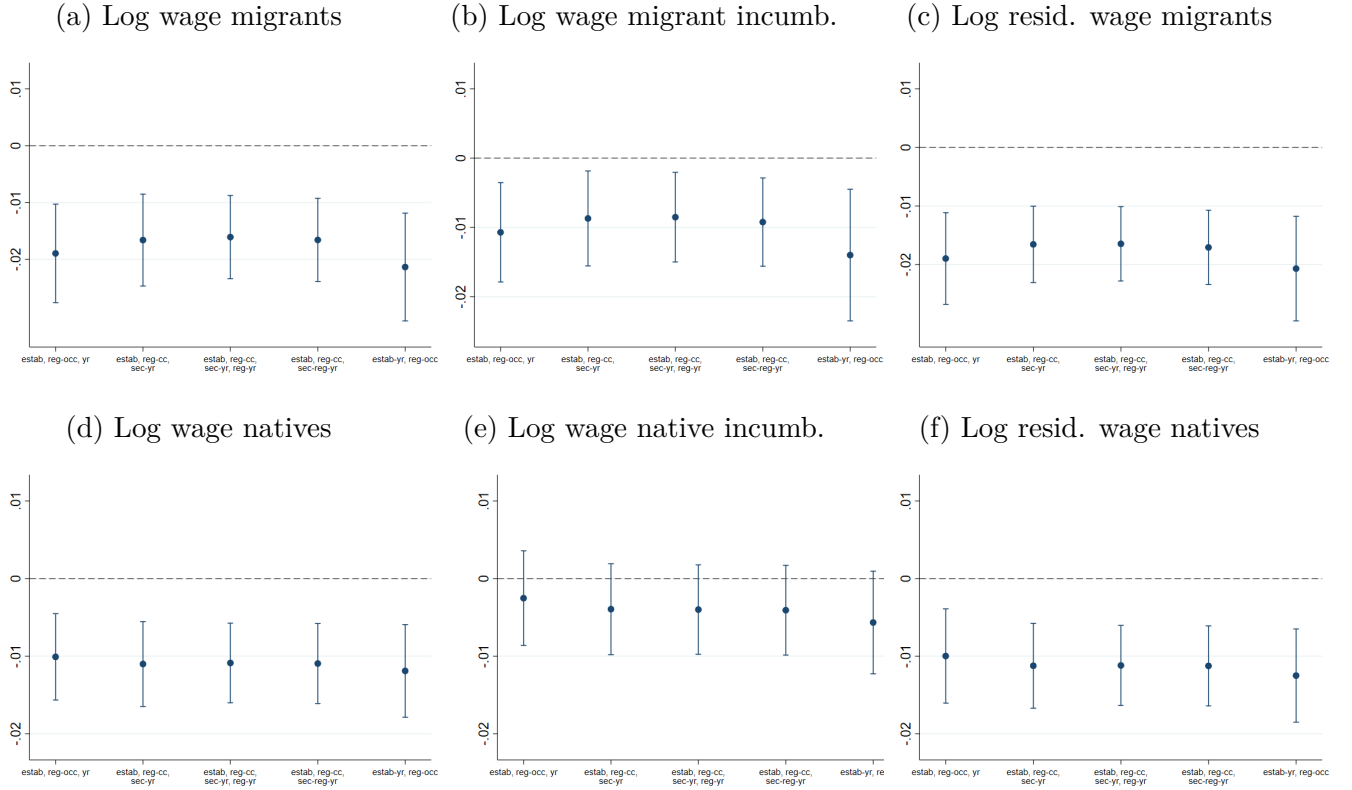
Note: The figure reports the average growth in profits within each of the major sectors in the sample, with the aim of showing that construction is the most badly hit by the financial crisis. The profit level in 2008 is normalized to 1 for all sectors. Data source : firm administrative financial records called *FICUS - FARE*.

Figure A12: Robustness tests - increasingly demanding fixed effects on employment outcomes



The figure reports the estimated effect of the policy across different specifications. The first dot from the left includes estab. FE, region x occupation FE and year FE. The second dot corresponds to the baseline specification, which adds sector x year FE. The third dot adds region x year FE, the fourth adds sector x region x year FE, and the fifth dot includes estab. x year FE, thus estimating the effect only within estab. employing both treatment and control occupations. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change relative to pre-reform mean, computed as the mean outcome observed within treated occupations before 2008.

Figure A13: Robustness tests - increasingly demanding fixed effects on wage outcomes



The figure reports the estimated effect of the policy across different specifications. The first dot from the left includes estab. FE, region x occupation FE and year FE. The second dot corresponds to the baseline specification, which adds sector x year FE. The third dot adds region x year FE, the fourth adds sector x region x year FE, and the fifth dot includes estab. x year FE, thus estimating the effect only within estab. employing both treatment and control occupations. Vertical bars represent the 95% confidence intervals. The y-axis can be interpreted in terms of % change, since coefficients are multiplied by 100.

A.2 Tables

Table A1: List of 30 Occupations part of the Reform

Occupation	Professional category	N. of Regions (total=22)
Executive of financial audit	Executive/Engineer	22
Computer scientist	Executive/Engineer	22
Senior computer scientist	Executive/Engineer	22
Draughtsman manager in electricity and electronics	Executive/Engineer	22
Sales technician	Technician/Foreman	20
Sales repr. for intermed. goods and raw materials	Technician/Foreman	15
Marchandiser (design of stores and shelves)	Technician/Foreman	18
Operations manager in insurance	Technician/Foreman	15
Mechanic of construction machinery and agricultural machinery	Technician/Foreman	21
Technician of industrial planning and methods	Technician/Foreman	14
Draughtsman manager mechanical construction	Technician/Foreman	13
Industrial draughtsman mechanic. construction and metallurgy	Technician/Foreman	13
Industrial draughtsman in electricity and electronics	Technician/Foreman	22
Production technician in mechanic. construction and metallurgy	Technician/Foreman	15
Quality-manager mechanic. construction and metallurgy	Technician/Foreman	18
Quality-manager in electricity and electronics	Technician/Foreman	5
Production technician of process industry	Technician/Foreman	11
Production technician of wood and furniture	Technician/Foreman	17
Installer and maintenance of elevators	Technician/Foreman	17
Compliance inspector	Technician/Foreman	13
Responsible for electronic maintenance	Technician/Foreman	5
Draughtsman of buildings and public work	Technician/Foreman	13
Construction surveyor	Technician/Foreman	10
Head of technical studies for buildings and public work	Technician/Foreman	21
Head of technical studies in construction	Technician/Foreman	1
Head of construction sites for buildings and public work	Technician/Foreman	21
Site supervisor for buildings and public work	Technician/Foreman	20
Operator of glass production	Blue collar	16
Installation driver for cement production	Blue collar	21
Driver of machinery for wood and furniture production	Blue collar	18
Artisan of wood and furniture production	Blue collar	18

Table A2: List of control occupations

Occupation	Rank	Occupation	Rank
Researchers in fundamental research	Executive/Engineer	Administrative and sales executives	Executive/Engineer
Executives in charge of market research	Executive/Engineer	Executive in charge of HR and training	Executive/Engineer
Technical directors	Executive/Engineer	Engineers in electricity and electronics	Executive/Engineer
Sales executives in electricity and electronics material	Executive/Engineer	Engineers in mechanics and metal work	Executive/Engineer
Sales executives in professional mechanical equipment	Executive/Engineer	Engineers in the process industry	Executive/Engineer
Sales executives in the process industry	Executive/Engineer	Engineers in energy and water distribution	Executive/Engineer
Engineers in paper, wood and soft materials	Executive/Engineer	Sales managers in industrial supplies	Executive/Engineer
Engineers in logistics	Executive/Engineer	Engineers in construction	Executive/Engineer
Engineers in production methods	Executive/Engineer	Engineers in the transport industry	Executive/Engineer
Engineers in tele-communications	Executive/Engineer	Photographers	Technician/Foreman
Buyers	Technician/Foreman	Technicians in industrial maintenance	Technician/Foreman
Foremen in the transport industry	Technician/Foreman	Intermediate professionals in the insurance industry	Technician/Foreman
Foremen in electricity and electronics	Technician/Foreman	Foremen in the food and chemicals industry	Technician/Foreman
Foremen in the metal and other process industry	Technician/Foreman	Foremen in electricity and electronics maintenance	Technician/Foreman
Foremen in the paper industry	Technician/Foreman	Foremen in mechanical maintenance	Technician/Foreman
Foremen in generic maintenance	Technician/Foreman	Concrete operators	Blue collar
Drivers of construction machinery	Blue collar	Installers of electricity and electronics	Blue collar
Blue collar workers in construction	Blue collar	Diggers in the extraction industry	Blue collar
Cable workers	Blue collar	Surveyors of electric and electronics equipment	Blue collar
Boilermakers	Blue collar	Pipefitters	Blue collar
Metal welders	Blue collar	Metal machinery operators	Blue collar
Mechanics assemblers	Blue collar	Metallic structures assemblers	Blue collar
Mechanics surveyors	Blue collar	Drivers of heavy machinery in the process industry	Blue collar
Blue collar workers in the chemicals and pharmaceutical industry	Blue collar	Blue collar workers in the paper and printing industry	Blue collar
Blue collar workers in the textile industry	Blue collar	Blue collar workers in plastic processing	Blue collar
Operators in meat transformation	Blue collar	Maintenance mechanics	Blue collar
Electromechanics	Blue collar	Qualified industrial machine settlers (metal and mechanics)	Blue collar
Qualified industrial machine settlers (other than metal and mechanics)	Blue collar	Qualified bricklayers	Blue collar
Carpenters	Blue collar	Roofers	Blue collar
Floor and carpet fitters	Blue collar	Interior joiners	Blue collar
Blue collar workers in construction maintenance	Blue collar	Electricians	Blue collar
Maintenance electricians	Blue collar	Moulders	Blue collar
Drivers of heavy machinery in warehouses	Blue collar	Locksmiths	Blue collar
Unskilled workers in transformation industry	Blue collar	Unskilled workers in printing industry	Blue collar

This list includes the occupations opened to Romanians and Bulgarians that are used as control group. In total the control group includes 99 categories of the PCS-4 digits classification. In this table I combined the very similar ones into a single line, for simplicity, which resulted in this list of 68 occupations.

Table A3: Summary statistics

	Treatment mn/(sd)	Control mn/(sd)	Alt. control mn/(sd)
Panel A : Stock variables			
Migrant empl.	1.2 (8.9)	1.4 (8.6)	1.3 (8.4)
Hourly wage migrants	23.5 (14.9)	21.1 (14.8)	21.6 (15.9)
Hourly wage migrant incumbents	23.9 (15.0)	21.3 (14.5)	21.9 (15.7)
Residualized log wage migrants	0.3 (0.4)	0.2 (0.5)	0.2 (0.5)
Native empl.	15.0 (51.2)	13.4 (42.0)	14.7 (45.0)
Hourly wage natives	23.1 (13.5)	23.4 (14.9)	23.4 (15.2)
Hourly wage native incumbents	23.6 (13.6)	23.8 (15.0)	23.7 (15.2)
Residualized log wage natives	0.22 (0.39)	0.19 (0.43)	0.20 (0.44)
Establishment size	159 (452)	167 (472)	166 (477)
Pre-ref average tightness	0.19 (0.20)	0.16 (0.12)	0.21 (0.13)
N. observations	1,603,081	1,936,186	2,512,118
N. occup x establishments	380,415	449,965	577,764
N. establishments	119,757	129,010	151,955
Panel B : Flow variables			
Migrant new hires	0.12 (1.66)	0.11 (1.39)	0.12 (1.40)
Prob. of hiring a migrant	0.05 (0.21)	0.05 (0.22)	0.05 (0.22)
Native new hires	2.08 (9.13)	1.97 (9.90)	2.30 (10.27)
Prob. of hiring a native	0.57 (0.49)	0.54 (0.50)	0.57 (0.50)
Estab. size	265 (661)	269 (671)	264 (676)
Pre-ref average tightness	0.19 (0.20)	0.17 (0.12)	0.22 (0.13)
N. observations	547,605	652,731	899,372
N. occup x establishments	248,118	303,907	409,969
N. establishments	87,621	97,557	126,334

Summary statistics of the sample used in the main regressions. Period: 2004 - 2013.

Table A4: Balancing test within regions

	Region - occup regressions				
	Constant		Treatment dummy		N. obs
Panel A: Stock variables					
Average levels					
Native empl.	21.46***	(0.792)	1.548*	(0.827)	2,014
Wage natives	22.26***	(0.423)	0.344	(0.371)	2,014
Residualized wage natives	0.0987***	(0.0164)	0.0265*	(0.0151)	2,014
Migrant empl.	2.477***	(0.114)	0.0816	(0.0877)	2,014
Wage migrants	22.57***	(0.474)	-0.128	(0.412)	1,923
Residualized wage migrants	0.185***	(0.0189)	0.00187	(0.0171)	1,923
Average yearly changes					
Native empl.	-0.643***	(0.151)	0.0129	(0.0890)	2,011
Wage natives	0.380***	(0.0885)	0.0323	(0.0554)	2,009
Residualized wage natives	0.000788	(0.00221)	0.000131	(0.00160)	2,010
Migrant empl.	-0.0343**	(0.0166)	-0.0272**	(0.0129)	2,010
Wage migrants	0.210*	(0.126)	0.152	(0.123)	1,890
Residualized wage migrants	0.000292	(0.00360)	0.00536	(0.00332)	1,890
Panel B: Flow variables					
Average levels					
N. native hires	2.578***	(0.139)	0.149	(0.0958)	2,057
P. hiring a native	0.586***	(0.00743)	0.0291***	(0.00724)	2,057
N. migrant hires	0.181***	(0.0138)	-0.0179**	(0.00787)	2,057
P. hiring a migrant	0.0781***	(0.00373)	-0.00331	(0.00269)	2,057
Average yearly changes					
N. native hires	0.0569	(0.0414)	-0.0321	(0.0333)	2,031
P. hiring a native	0.00782	(0.00587)	8.94e-05	(0.00498)	2,031
N. migrant hires	-0.0178**	(0.00785)	0.0129***	(0.00493)	2,031
P. hiring a migrant	0.00184	(0.00215)	0.00156	(0.00153)	2,03

Each line corresponds to a separate regression where the outcomes averaged at the region-occupation level are regressed on the treatment dummy over the period preceding the reform (2004-07), controlling for the propensity score. Wages of incumbents are not reported because identical to the total wages in this period (the sample restriction only applies after the reform). In addition to the level of outcomes, I also report the effect on the yearly change. Standard errors clustered at the region x occupation level. The constant term can be interpreted as the control average conditional on the propensity score.

Table A5: Balancing test within establishments using the alternative controls

	Estab - occup regressions				
	Constant		Treatment dummy		N. obs
Panel A: Stock variables					
Average levels					
Native empl.	18.10***	(0.981)	1.802*	(1.090)	468,973
Wage natives	29.75***	(0.574)	0.411	(0.630)	455,635
Residualized wage natives	0.386***	(0.0190)	0.0268	(0.0189)	455,804
Migrant empl.	2.144***	(0.196)	-0.123	(0.198)	469,025
Wage migrants	26.30***	(0.772)	1.334	(0.946)	94,122
Residualized wage migrants	0.343***	(0.0269)	0.0704**	(0.0288)	94,168
Average yearly changes					
Native empl.	-0.685***	(0.0763)	-0.0744	(0.100)	350,728
Wage natives	0.443***	(0.0372)	0.0395	(0.0349)	339,295
Residualized wage natives	0.000583	(0.000795)	0.000965	(0.000800)	339,998
Migrant empl.	-0.143***	(0.0201)	-0.00786	(0.0202)	350,931
Wage migrants	0.638***	(0.110)	0.146	(0.127)	59,971
Residualized wage migrants	0.00849***	(0.00245)	-0.00227	(0.00283)	60,096
Panel B: Flow variables					
Average levels					
N. native hires	2.342***	(0.191)	0.204	(0.175)	278,470
P. hiring a native	0.553***	(0.00520)	0.0187***	(0.00518)	278,470
N. migrant hires	0.156***	(0.0238)	0.00832	(0.0237)	278,470
P. hiring a migrant	0.0634***	(0.00408)	0.000629	(0.00402)	278,470
Average yearly changes					
N. native hires	-0.371***	(0.0933)	-0.0493	(0.0721)	93,009
P. hiring a native	-0.0261***	(0.00498)	-4.82e-05	(0.00530)	93,117
N. migrant hires	-0.0458***	(0.00895)	0.00834	(0.00937)	92,992
P. hiring a migrant	0.00861***	(0.00185)	0.000750	(0.00188)	93,117

Each line corresponds to a separate regression where the outcomes are regressed on the alternative treatment dummy over the period preceding the reform (2004-07), controlling for the propensity score and establishment fixed effects. Wages of incumbents are not reported because identical to the total wages in this period (the sample restriction only applies after the reform). In addition to the level of outcomes, I also report the effect on the yearly change. standard errors clustered at the region x occupation level. The constant term can be interpreted as the control average conditional on the fixed effects and the propensity score.

Table A6: Balancing test within regions using the alternative controls

	Region - occup regressions				
	Constant		Treatment dummy		N. obs
Panel A: Stock variables					
Average levels					
Native empl.	18.59***	(0.786)	2.726***	(0.715)	2,699
Wage natives	24.76***	(0.419)	-0.594*	(0.318)	2,698
Residualized wage natives	0.190***	(0.0161)	-0.0385***	(0.0131)	2,698
Migrant empl.	1.894***	(0.0928)	0.373***	(0.0747)	2,699
Wage migrants	24.53***	(0.497)	-0.784**	(0.371)	2,367
Residualized wage migrants	0.259***	(0.0187)	-0.0444***	(0.0154)	2,367
Average yearly changes					
Native empl.	-0.610***	(0.176)	0.0622	(0.0841)	2,670
Wage natives	0.411***	(0.0818)	0.0296	(0.0566)	2,668
Residualized wage natives	-0.00514	(0.00522)	0.00504**	(0.00228)	2,669
Migrant empl.	-0.0502***	(0.0151)	-0.0231*	(0.0125)	2,668
Wage migrants	0.199	(0.126)	0.150	(0.116)	2,282
Residualized wage migrants	-0.000442	(0.00374)	0.00480	(0.00364)	2,282
Panel B: Flow variables					
Average levels					
N. native hires	2.583***	(0.160)	-0.122	(0.101)	2,741
P. hiring a native	0.593***	(0.00752)	0.0130*	(0.00709)	2,741
N. migrant hires	0.147***	(0.0123)	-0.00737	(0.00658)	2,741
P. hiring a migrant	0.0635***	(0.00326)	0.000382	(0.00257)	2,741
Average yearly changes					
N. native hires	0.00680	(0.0643)	0.0360	(0.0428)	2,641
P. hiring a native	-0.00768	(0.00663)	0.00701	(0.00602)	2,643
N. migrant hires	-0.00775	(0.00693)	0.00643	(0.00464)	2,642
P. hiring a migrant	0.00194	(0.00239)	0.00455**	(0.00190)	2,642

Each line corresponds to a separate regression where the outcomes averaged at the region-occupation level are regressed on the alternative treatment dummy over the period preceding the reform (2004-07), controlling for the propensity score. Wages of incumbents are not reported because identical to the total wages in this period (the sample restriction only applies after the reform). In addition to the level of outcomes, I also report the effect on the yearly change. Standard errors clustered at the region x occupation level. The constant term can be interpreted as the control average conditional on the propensity score.

Table A7: Balancing test within establishments on demographics

	Treat			Estab - occup regressions			Treat-alt	
	Constant	Treatment dummy	N. obs	Constant	Treatment dummy	N. obs		
Panel A: Stock variables								
Average levels								
Age natives	41.77*** (0.225)	-0.907*** (0.232)	395,651	41.70*** (0.190)	-1.002*** (0.226)	455,804		
Sh. Women natives	0.105*** (0.00721)	0.0310*** (0.00921)	395,651	0.0951*** (0.00669)	0.00352 (0.00993)	455,804		
Experience natives	1.599*** (0.00465)	0.00102 (0.00509)	395,651	1.593*** (0.00417)	0.00641 (0.00529)	455,804		
Age migrants	41.12*** (0.309)	-0.445 (0.313)	82,232	40.84*** (0.244)	-0.445 (0.272)	94,168		
Sh. women migrants	0.121*** (0.00839)	0.0284** (0.0113)	82,232	0.121*** (0.00828)	0.00642 (0.0124)	94,168		
Experience migrants	1.444*** (0.0135)	-0.00572 (0.0135)	82,232	1.434*** (0.0118)	-0.00348 (0.0129)	94,168		
Average yearly changes								
Age natives	0.533*** (0.0155)	0.00163 (0.0143)	293,529	0.521*** (0.0143)	0.0130 (0.0142)	340,204		
Sh. Women natives	0.00138*** (0.000459)	0.00150*** (0.000398)	293,825	0.00171*** (0.000417)	0.000858** (0.000397)	340,519		
Experience natives	0.922*** (0.00227)	-0.00280 (0.00208)	293,825	0.919*** (0.00209)	-0.00344 (0.00218)	340,519		
Age migrants	0.484*** (0.0497)	0.00968 (0.0514)	52,913	0.522*** (0.0459)	0.00909 (0.0522)	60,153		
Sh. women migrants	-0.000999 (0.00129)	0.000748 (0.00168)	52,944	0.000372 (0.00135)	0.000643 (0.00176)	60,189		
Experience migrants	0.873*** (0.00691)	0.0159** (0.00731)	52,944	0.874*** (0.00635)	0.0126* (0.00754)	60,189		
Panel B: Flow variables								
Average levels								
Age natives	33.64*** (0.263)	-0.561** (0.266)	144,128	34.03*** (0.220)	-0.920*** (0.256)	171,452		
Sh. Women natives	0.126*** (0.00792)	0.0379*** (0.00967)	144,128	0.133*** (0.00706)	0.00334 (0.00993)	171,452		
Age migrants	33.77*** (0.554)	-0.703 (0.583)	13,639	33.92*** (0.454)	-0.757 (0.529)	16,175		
Sh. women migrants	0.152*** (0.0219)	0.0385 (0.0292)	13,639	0.163*** (0.0205)	0.0202 (0.0270)	16,175		
Average yearly changes								
Age natives	0.250* (0.143)	-0.101 (0.140)	37,480	0.294** (0.119)	-0.114 (0.128)	46,462		
Sh. Women natives	0.00446 (0.00599)	0.00708 (0.00564)	37,480	0.00630 (0.00508)	0.00635 (0.00526)	46,462		
Age migrants	0.653 (1.455)	-0.821 (1.299)	2,131	0.131 (1.218)	-0.638 (1.126)	2,593		
Sh. women migrants	-0.00554 (0.0663)	-0.0797 (0.0678)	2,131	0.00720 (0.0666)	-0.0621 (0.0637)	2,593		

Each line corresponds to a separate regression where the outcomes are regressed on the treatment dummy over the period preceding the reform (2004-07), controlling for the propensity score and establishment fixed effects. Wages of incumbents are not reported because identical to the total wages in this period (the sample restriction only applies after the reform). In addition to the level of outcomes, I also report the effect on the yearly change. standard errors clustered at the region x occupation level. The constant term can be interpreted as the control average conditional on the fixed effects and the propensity score.

Table A8: Balancing test within regions on demographics

	Region - occup regressions				Treat-alt	
	Constant	Treatment dummy	N. obs	Constant	Treatment dummy	N. obs
Panel A: Stock variables						
Average levels						
Age natives	40.03*** (0.226)	-1.365*** (0.00606)	2,014	40.66*** (0.215)	-1.110*** (0.190)	2,698
Sh. Women natives	0.0667*** (0.00629)	0.00606 (0.00700)	2,014	0.0555*** (0.00802)	-0.142*** (0.00918)	2,698
Experience natives	1.575*** (0.00882)	-0.00441 (0.00792)	2,014	1.587*** (0.0104)	0.0423*** (0.00921)	2,698
Age migrants	41.43*** (0.251)	-1.130*** (0.228)	1,923	40.98*** (0.253)	-0.468** (0.236)	2,367
Sh. women migrants	0.0747*** (0.00753)	0.000405 (0.00806)	1,923	0.0715*** (0.00944)	-0.109*** (0.0102)	2,367
Experience migrants	1.471*** (0.0154)	0.00709 (0.0144)	1,923	1.455*** (0.0162)	0.0794*** (0.0164)	2,367
Average yearly changes						
Age natives	0.127*** (0.0437)	-0.000276 (0.0372)	2,009	0.228*** (0.0494)	-0.0734 (0.0447)	2,667
Sh. Women natives	0.000364 (0.000910)	0.000872 (0.000882)	2,009	0.000982 (0.00165)	0.00162 (0.00164)	2,665
Experience natives	0.800*** (0.00640)	-0.00571 (0.00516)	2,010	0.815*** (0.00736)	0.00876 (0.00558)	2,669
Age migrants	-0.262** (0.105)	-0.214* (0.111)	1,889	-0.0635 (0.0983)	-0.239** (0.109)	2,280
Sh. women migrants	0.00852*** (0.00267)	0.000892 (0.00287)	1,888	0.00822*** (0.00310)	0.00364 (0.00332)	2,279
Experience migrants	0.759*** (0.0105)	-0.0114 (0.00941)	1,890	0.756*** (0.0107)	0.00314 (0.0102)	2,282
Panel B: Flow variables						
Average levels						
Age natives	31.90*** (0.218)	-1.291*** (0.200)	2,038	32.84*** (0.216)	-1.454*** (0.195)	2,688
Sh. Women natives	0.0830*** (0.00701)	0.00173 (0.00777)	2,038	0.101*** (0.00872)	-0.139*** (0.00945)	2,688
Age migrants	33.55*** (0.368)	-1.764*** (0.351)	1,550	33.48*** (0.329)	-1.039*** (0.328)	1,803
Sh. women migrants	0.0531*** (0.0137)	-0.00165 (0.0150)	1,550	0.0768*** (0.0132)	-0.0554*** (0.0151)	1,803
Average yearly changes						
Age natives	0.231* (0.136)	-0.0661 (0.129)	1,993	0.147 (0.134)	-0.182 (0.135)	2,544
Sh. Women natives	0.00427 (0.00473)	0.00158 (0.00509)	1,994	0.0138** (0.00561)	-0.000487 (0.00540)	2,545
Age migrants	-0.463 (0.484)	0.654 (0.511)	961	-0.802* (0.434)	0.484 (0.459)	1,094
Sh. women migrants	-0.0131 (0.0172)	-0.0242 (0.0192)	961	-0.00144 (0.0164)	-0.0252 (0.0205)	1,094

Each line corresponds to a separate regression where the outcomes averaged at the region-occupation level are regressed on the treatment dummy over the period preceding the reform (2004-07), controlling for the propensity score. Wages of incumbents are not reported because identical to the total wages in this period (the sample restriction only applies after the reform). In addition to the level of outcomes, I also report the effect on the yearly change. Standard errors clustered at the region x occupation level. The constant term can be interpreted as the control average conditional on the propensity score.

Table A9: Balancing test within establishments without controlling for the propensity score

	Estab - occup regressions				
	Constant		Treatment dummy		N. obs
Panel A: Stock variables					
Average levels					
Native empl.	13.33***	(0.438)	0.498	(0.839)	424,624
Wage natives	23.94***	(0.483)	-1.387*	(0.714)	411,949
Residualized wage natives	0.217***	(0.0135)	-0.0108	(0.0208)	412,111
Migrant empl.	1.480***	(0.118)	-0.466***	(0.179)	424,677
Wage migrants	21.42***	(0.567)	0.493	(0.984)	84,582
Residualized wage migrants	0.195***	(0.0158)	0.0588**	(0.0298)	84,628
Average yearly changes					
Native empl.	-0.445***	(0.0387)	-0.0328	(0.0743)	315,948
Wage natives	0.593***	(0.0213)	0.0555*	(0.0327)	305,149
Residualized wage natives	0.00596***	(0.000430)	0.00181**	(0.000710)	305,770
Migrant empl.	-0.0839***	(0.0119)	0.0133	(0.0177)	316,128
Wage migrants	0.758***	(0.0553)	0.114	(0.117)	54,249
Residualized wage migrants	0.0140***	(0.00135)	0.000297	(0.00273)	54,356
Panel B: Flow variables					
Average levels					
N. native hires	1.580***	(0.0733)	0.0483	(0.142)	238,947
P. hiring a native	0.527***	(0.00326)	0.0256***	(0.00488)	238,947
N. migrant hires	0.0946***	(0.00765)	-0.0236	(0.0181)	238,947
P. hiring a migrant	0.0436***	(0.00196)	-0.00968***	(0.00375)	238,947
Average yearly changes					
N. native hires	-0.303***	(0.0465)	0.0264	(0.0609)	78,283
P. hiring a native	-0.0392***	(0.00307)	-0.00178	(0.00533)	78,377
N. migrant hires	-0.0370***	(0.00610)	0.0247***	(0.00914)	78,276
P. hiring a migrant	0.00442***	(0.00118)	0.000952	(0.00183)	78,377

Each line corresponds to a separate regression where the outcomes are regressed on the treatment dummy over the period preceeding the reform (2004-07), controlling for establishment fixed effects. Wages of incumbents are not reported because identical to the total wages in this period (the sample restriction only applies after the reform). In addition to the level of outcomes, I also report the effect on the yearly change. standard errors clustered at the region x occupation level. The constant term can be interpreted as the control average conditional on the fixed effects.

Table A10: Main Results on Employment using the alternative controls

	(1)	(2)	(3)	(4)	(5)	(6)
	N. hires		P. hiring		Employment	
	Migrants	Natives	Migrants	Natives	Migrants	Natives
Panel A : with estab. fixed effects						
treat_alt x post reform	0.0405** (0.0202)	0.0615 (0.0551)	0.00471*** (0.00159)	0.00201 (0.00271)	0.215*** (0.0782)	0.770** (0.326)
Baseline trt mean	0.0938	2.089	0.0436	0.582	1.252	16.12
Effect in % growth	43%	3%	11%	0%	17%	5%
Observations	1,387,094	1,387,094	1,387,094	1,387,094	3,923,203	3,922,668
R-squared	0.216	0.304	0.203	0.209	0.254	0.257
Panel B : without estab. fixed effects						
treat_alt x post reform	0.0447* (0.0249)	0.0333 (0.0444)	0.00483** (0.00192)	0.00249 (0.00304)	0.188** (0.0750)	0.536* (0.318)
Baseline trt mean	0.0938	2.089	0.0436	0.582	1.252	16.12
Effect in % growth	48%	2%	11%	0%	15%	3%
Observations	1,424,335	1,424,335	1,424,335	1,424,335	3,932,155	3,931,621
R-squared	0.033	0.088	0.067	0.050	0.052	0.071

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. Treat_alt represents the treatment definition where controls are occupations in the top 2 quantiles of tightness. Migrants refers to individuals born outside of France for the stock variables and to individuals born outside the EU for the flow variables. Natives refers to individuals born in France. Baseline mean reports the mean value of the outcome for treated occupations prior to the reform, and serves to compute the effect in terms of growth rate relative to baseline.

Table A11: Flow outcomes: Europeans vs non-Europeans

	(1)	(2)	(3)	(4)	(5)	(6)
	Migrants non-EU		Migrants EU		All migrants	
	N. hires	P. hiring	N. hires	P. hiring	N. hires	P. hiring
Panel A : with estab. fixed effects						
treat x post reform	0.0483** (0.0222)	0.00693*** (0.00169)	0.00811*** (0.00296)	0.00187** (0.000866)	0.0564** (0.0240)	0.00589*** (0.00172)
Baseline mean	0.0938	0.0436	0.0307	0.0215	0.124	0.0589
Effect in % growth	51%	16%	26%	9%	45%	10%
Observations	1,162,063	1,162,063	1,162,063	1,162,063	1,162,063	1,162,063
Pseudo R-squared	0.214	0.195	0.205	0.175	0.223	0.211
Panel B : without estab. fixed effects						
treat x post reform	0.0523** (0.0259)	0.00670*** (0.00202)	0.00671** (0.00308)	0.00172* (0.000971)	0.0590** (0.0277)	0.00572*** (0.00204)
Baseline mean	0.0938	0.0436	0.0307	0.0215	0.124	0.0589
Effect in % growth	56%	15%	22%	8%	48%	10%
Observations	1,194,530	1,194,530	1,194,530	1,194,530	1,194,530	1,194,530
Pseudo R-squared	0.032	0.066	0.039	0.052	0.038	0.079

Standard errors clustered at the region-occupation level. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. Non-EU migrants refers to individuals born outside the European Union, EU Migrants refers to individuals born in the EU, and all migrants include all individuals born outside of France. Baseline mean reports the mean value of the outcome for treated occupations prior to the reform, and serves to compute the effect in terms of growth rate relative to baseline.

Table A12: Main results on wages using the alternative controls

	(1) log wage	(2) log wage	(3) log wage incumbents	(4) log wage incumbents	(5) log wage residualized	(6) log wage residualized
	Migrants	Natives	Migrants	Natives	Migrants	Natives
Panel A : with estab. fixed effects						
treat_alt x post reform	-0.0143*** (0.00385)	-0.00904*** (0.00249)	-0.00667** (0.00330)	-0.00260 (0.00266)	-0.0152*** (0.00305)	-0.00849*** (0.00241)
D Mig - Nat.	-0.00531*** (0.000190)		-0.00407*** (0.000162)		-0.00668*** (0.000152)	
Observations	674,211	3,789,310	597,213	3,508,091	674,508	3,790,987
R-squared	0.739	0.671	0.772	0.693	0.750	0.675
Panel B : without estab. fixed effects						
treat_alt x post reform	-0.0219*** (0.00597)	-0.0130*** (0.00348)	-0.0125** (0.00535)	-0.00597* (0.00336)	-0.0203*** (0.00511)	-0.0108*** (0.00345)
D Mig - Nat.	-0.00883*** (0.000469)		-0.00648*** (0.000350)		-0.00952*** (0.000358)	
Observations	690,509	3,798,625	612,182	3,517,546	690,807	3,800,300
R-squared	0.541	0.540	0.569	0.554	0.565	0.545

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. Treat_alt represents the treatment definition where controls are occupations in the top 2 quantiles of tightness. Migrants refers to individuals born outside of France, while natives refers to individuals born in France. Incumbents are individuals already working in France before 2008, while residualized wages control for age, gender and experience of workers within the establishment. D Mig - Nat computes the difference in the wage effect between migrants and natives, and report whether it is significant.

Table A13: Effect of reforms on workers' wage trajectories

	(1)	(2)	(3)	(4)
	asinh wage following workers (also out of employment)			
	all estab.		only estab. > 20	
	Migrants	Natives	Migrants	Natives
Panel A : with firm fixed effects				
treat x post reform	-0.00728 (0.0156)	0.00811 (0.0163)	0.00933 (0.0158)	0.0263** (0.0123)
D Mig - Nat.		-0.0154 (0.0122)		-0.0170 (0.0111)
Observations	429,278	2,658,254	262,000	1,465,093
R-squared	0.438	0.410	0.463	0.410
Panel B : without firm fixed effects				
treat x post reform	-0.00735 (0.0145)	0.00810 (0.0156)	0.00927 (0.0149)	0.0263** (0.0120)
D Mig - Nat.		-0.0154 (0.0159)		-0.0170 (0.0154)
Observations	429,278	2,658,256	262,000	1,465,094
R-squared	0.152	0.155	0.162	0.139

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1
Standard errors are clustered at the occupation x region level. Treatment is assigned to individuals in 2006 according to their occupation at the time. Wage outcomes follow individuals up to 2010 regardless of whether they remain or not in the occupation, assigning a wage of 0 to workers that exit the labor force. Regressions include the same controls as in the main analysis: the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects (all defined based on the worker location in 2006). Panel B also includes establishment fixed effects. Columns (3) and (4) only consider individuals that in 2006 work in establishments with more than 20 employees. Period of analysis: 2006 - 2010.

Table A14: Wage elasticity using the alternative controls

	(1)	(2)	(3)	(4)	(5)
	Log mig empl	Log wage Migrants	Natives	Log wage incumbent sample Migrants	Natives
	First stage	IV	IV	IV	IV
Panel A: with estab fixed effects					
treat_alt x post reform	0.119*** (0.0216)				
Log mig empl		-0.137*** (0.0386)	-0.0819*** (0.0197)	-0.0675** (0.0311)	-0.0260 (0.0219)
Observations	3,890,930	674,211	3,756,821	597,213	3,478,654
R-squared / RMSE	0.984	0.261	0.271	0.241	0.257
K-P F statistic		14.97	29.76	13.98	29.08
Panel B: without estab fixed effects					
treat_alt x post reform	0.125*** (0.0241)				
Log mig empl		-0.188*** (0.0439)	-0.110*** (0.0230)	-0.112*** (0.0394)	-0.0530** (0.0242)
Observations	3,899,983	690,509	3,766,240	612,182	3,488,197
R-squared / RMSE	0.983	0.328	0.314	0.313	0.303
K-P F statistic		14.94	26.70	14.43	26.41

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. The table presents the IV results where the log of migrant employment in a given occupation x region is instrumented using the interaction between the alternative treatment indicator and the post-reform period. The outcome wages are measured within occupations and establishments. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. The bottom of the page reports the Kleibergen-Paap F-statistic.

Table A15: Wage elasticity to change in total employment in the estab.

	(1)	(2)	(3)	(4)	(5)
	Log empl	Log wage Migrants	Natives	Log wage incumbent sample Migrants	Natives
	First stage	IV	IV	IV	IV
Panel A: with estab fixed effects					
treat x post reform	0.0432*** (0.0117)				
Log empl		-0.450*** (0.145)	-0.304*** (0.0689)	-0.245*** (0.0898)	-0.112** (0.0478)
Observations	3,386,243	591,466	3,264,116	525,758	3,020,198
R-squared	0.339	-2.694	-1.166	-0.731	-0.128
RMSE		0.492	0.398	0.313	0.273
K-P F statistic		7.776	14.78	9.067	29.84
Panel B: without estab fixed effects					
treat x post reform	0.0357*** (0.0132)				
Log empl		-0.508*** (0.164)	-0.423*** (0.112)	-0.244*** (0.0751)	-0.159*** (0.0549)
Observations	3,396,412	604,851	3,274,545	537,974	3,030,299
R-squared / RMSE	0.106	0.735	0.574	0.436	0.353
K-P F statistic		7.002	8.277	13.17	21.23

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. The table presents the IV results where the log of employment in a given establishment x region is instrumented using the interaction between the treatment indicator and the post-reform period. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. The bottom of the page reports the Kleibergen-Paap F-statistic.

Table A16: Effect on age, gender and experience composition

	(1)	(2)	(3)	(4)	(5)	(6)
		Migrants			Natives	
VARIABLES	average age	share women	average experience	average age	share women	average experience
Panel A : Hiring Flows						
treat x post reform	-0.329 (0.231)	0.0190* (0.0108)	- -	0.0375 (0.0612)	0.00576** (0.00229)	- -
Baseline trt mean	31.70	0.224	-	31.12	0.202	-
Effect in % growth	-1%	8%	-	0%	3%	-
Observations	44,866	44,866	-	637,135	637,135	-
R-squared	0.476	0.494	-	0.329	0.331	-
Panel B : Employment Stock						
treat x post reform	-0.155** (0.0719)	0.00335 (0.00263)	-0.0422 (0.0275)	-0.107** (0.0481)	-0.00245 (0.00153)	-0.0980*** (0.0192)
Baseline trt mean	40.09	0.169	1.203	39.68	0.168	1.258
Effect in % growth	0%	2%	-4%	0%	-1%	-8%
Observations	609,812	609,812	609,812	3,403,892	3,403,892	3,403,892
R-squared	0.541	0.578	0.781	0.362	0.397	0.789
Panel C : Employment Stock of incumbents						
treat x post reform	-0.0329 (0.0707)	0.00235 (0.00263)	-0.0124* (0.00711)	0.0643 (0.0540)	-0.00334** (0.00140)	-0.0214*** (0.00504)
Baseline trt mean	40.09	0.169	1.203	39.68	0.168	1.258
Effect in % growth	0%	1%	-1%	0%	-2%	-2%
Observations	542,296	542,296	542,296	3,149,490	3,149,490	3,149,490
R-squared	0.567	0.612	0.960	0.366	0.410	0.956

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. All regressions control for estab. fixed effects, the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A looks at the composition within new hires, Panel B does the same for the composition of employment, and Panel C restricts employment to workers already present in France prior to 2008. Period of analysis: 2004-2013. Migrants refers to individuals born outside of France for the stock variables and to individuals born outside the EU for the flow variables. Baseline mean reports the mean value of the outcome for treated occupations prior to the reform, and serves to compute the effect in terms of growth rate relative to baseline.

Table A17: Effect on employer and occupation mobility

	(1)	(2)	(3)	(4)
	P. switching estab.		P. switch estab. and occup.	
	Migrants	Natives	Migrants	Natives
Panel A : with estab. fixed effects				
treat x post reform	0.00518* (0.00314)	0.00222 (0.00175)	0.00316 (0.00217)	0.00249* (0.00138)
Baseline trt mean	0.150	0.134	0.0910	0.0885
Effect in % growth	3%	2%	3%	3%
Observations	468,622	2,402,156	443,941	2,315,253
R-squared	0.489	0.508	0.505	0.525
Panel B : without estab. fixed effects				
treat x post reform	-0.000309 (0.00265)	-0.000436 (0.00109)	-0.000296 (0.00216)	-2.97e-05 (0.00103)
Baseline trt mean	0.150	0.134	0.0910	0.0885
Effect in % growth	0%	0%	0%	0%
Observations	483,334	2,415,744	458,572	2,329,581
R-squared	0.240	0.296	0.297	0.361

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes establishment fixed effects while panel B does not. Period of analysis: 2004-2013. Migrants refers to individuals born outside of France for the stock variables and to individuals born outside the EU for the flow variables. Baseline mean reports the mean value of the outcome for treated occupations prior to the reform, and serves to compute the effect in terms of growth rate relative to baseline.

Table A18: Effects within estab. size, productivity and share of migrant quartiles

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	log wage		log wage incumbents		log wage residualized	
	Migrants	Natives	Migrants	Natives	Migrants	Natives
Panel A : effect within quartiles of estab. size						
treat x post reform	-0.0162*** (0.00398)	-0.0108*** (0.00275)	-0.00861** (0.00346)	-0.00392 (0.00298)	-0.0166*** (0.00334)	-0.0112*** (0.00279)
D Mig - Nat.	-0.00538*** (0.000178)		-0.00468*** (0.000171)		-0.00537*** (0.000146)	
Observations	608,992	3,401,571	541,654	3,147,475	609,223	3,402,985
R-squared	0.723	0.658	0.754	0.677	0.733	0.659
Panel B : effect within quartiles of estab. productivity						
treat x post reform	-0.0166*** (0.00403)	-0.0112*** (0.00277)	-0.00840** (0.00339)	-0.00352 (0.00297)	-0.0170*** (0.00320)	-0.0113*** (0.00277)
D Mig - Nat.	-0.00544*** (0.000169)		-0.00487*** (0.000168)		-0.00566*** (0.000147)	
Observations	579,338	3,256,325	516,503	3,016,765	579,555	3,257,677
R-squared	0.720	0.655	0.751	0.674	0.730	0.656
Panel C : effect within quartiles of estab. shares of migrants						
treat x post reform	-0.0167*** (0.00401)	-0.0109*** (0.00277)	-0.00897*** (0.00341)	-0.00384 (0.00299)	-0.0165*** (0.00327)	-0.0111*** (0.00278)
D Mig - Nat.	-0.00592*** (0.000173)		-0.00518*** (0.000174)		-0.00538*** (0.000143)	
Observations	596,148	3,331,761	531,294	3,087,822	596,375	3,333,152
R-squared	0.722	0.658	0.753	0.677	0.732	0.659

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Panel A includes fixed effects for quartiles of firm size interacted with year dummies. Panel B includes fixed effects for quartiles of firm productivity interacted with year dummies. Panel C includes fixed effects for quartiles of firm share of migrant employment in all other occupations interacted with year dummies. Period of analysis: 2004-2013. Migrants refers to individuals born outside of France, while natives refers to individuals born in France. Incumbents are individuals already working in France before 2008, while residualized wages control for age, gender and experience of workers within the establishment. D Mig - Nat computes the difference in the wage effect between migrants and natives, and report whether it is significant.

Table A19: Elasticity of substitution between migrants and natives obtained from OLS

VARIABLES	(1)	(2)	(3)	(4)
	Log wage		Log incumbent wage	
	Migrants	Natives	Migrants	Natives
	OLS	OLS	OLS	OLS
Log ratio migrant to native empl.	-0.0394*** (0.00383)	-0.0183*** (0.00191)	-0.0314*** (0.00334)	-0.0177*** (0.00196)
D Mig - Nat. $-(\frac{1}{\sigma} + \Delta\phi_{ot,1})$	-0.0211*** (0.000240)		-0.0137*** (0.000186)	
Observations	589,021	3,247,656	523,685	3,005,526
R2	0.727	0.663	0.758	0.682

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. The table presents the OLS results where the log ratio of migrant employment over native employment in a given occupation x region is regressed on the wages of natives and migrants measured within occupations and establishments. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. The table also reports the difference between the migrant and native wage effects and the implied parameter capturing the elasticity of substitution + the difference in bargaining power between migrants and natives.

Table A20: Elasticity of substitution between migrants and natives in firms hiring both types

VARIABLES	(1)	(2) (3)		(4)	(5)
	Log ratio mig to native empl.	Log wage		Log incumbent wage	
		Migrants	Natives	Migrants	Natives
	First stage	IV	IV	IV	IV
treat x post reform	0.0377** (0.0182)				
Log ratio migrant to native empl.		-0.405** (0.193)	-0.289** (0.121)	-0.225* (0.127)	-0.152* (0.0806)
D Mig - Nat. $-(\frac{1}{\sigma} + \Delta\phi_{ot,1})$		-0.116 (0.299)		-0.0722 (0.180)	
Observations	786,142	514,436	740,210	465,064	712,636
R2 / RMSE	0.970	0.252	0.218	0.233	0.209
K-P F-statistic		3.523	4.255	3.074	4.120

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Standard errors clustered at the region-occupation level. The table presents the IV results where the log ratio of migrant employment over native employment in a given occupation x region is instrumented using the interaction between the treatment indicator and the post-reform period. The outcome wages are measured within occupations and establishments. The sample is restricted to firms hiring both migrants and natives over the period. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. Column (1) shows the first stage regression. Columns (2) and (3) present the IV estimation on migrant and native wages, and columns (4) and (5) present the IV estimation on incumbent migrant and native wages. The bottom of the page reports the Kleibergen-Paap F-statistic for the instrument relevance. The table also reports the difference between the migrant and native wage effects and the implied parameter capturing the elasticity of substitution + the difference in bargaining power between migrants and natives.

Table A21: Heterogeneity in the elasticity of substitution between migrants and natives

VARIABLES	(1)	(2)	(3)	Log wage		(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Low monopsony power		Migrants	Natives	Migrants	Natives	High job to job mobility		Low monopsony power		Log incumbent wage		Natives
	Migrants	Natives	Migrants	Natives	Migrants	Natives	Migrants	Natives	Migrants	Natives	Migrants	Natives	Natives
Log ratio migrant to native empl.	-0.446** (0.219)	-0.111 (0.0966)	-0.577** (0.293)	-0.400** (0.194)	-0.331** (0.142)	-0.426 (0.315)	-0.259* (0.151)	0.0246 (0.113)	-0.408* (0.214)	-0.199* (0.120)	-0.180** (0.0899)	-0.168 (0.180)	
D Mig - Nat. $-(\frac{1}{\sigma} + \Delta\phi_{n,t})$	-0.335 (0.257)		-0.177 (0.424)		0.0949 (1.554)		-0.284 (0.235)		-0.209 (0.282)		-0.0124 (0.534)		
Observations	183,574	1,190,046	331,768	1,646,853	351,720	1,644,081	172,998	1,168,685	296,336	1,525,920	308,591	1,500,887	
RMSE	0.256	0.266	0.256	0.269	0.259	0.282	0.245	0.263	0.236	0.252	0.238	0.259	
K-P F-statistic	3.265	2.699	3.216	3.798	4.508	1.497	2.867	2.668	2.809	3.745	4.179	1.497	

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1
Standard errors clustered at the region-occupation level. The table presents the IV results where the log ratio of migrant employment over native employment in a given occupation x region is instrumented using the interaction between the treatment indicator and the post-reform period. The outcome wages are measured within occupations and establishments. All regressions control for the propensity score interacted with year fixed effects, occupation x region fixed effects, and sector x time fixed effects. The bottom of the page reports the Kleibergen-Paap F-statistic for the instrument relevance. The table also reports the difference between the migrant and native wage effects and the implied parameter capturing the elasticity of substitution + the difference in bargaining power between migrants and natives. The various columns present the IV estimates for different sub-samples : The 50% of establishments with the lowest monopsony power, measured in terms of the share of the total region x occupation employment concentrated within the estab. (Columns 1, 2, 7 and 8); the 50% of regions x occupations with the lowest employer concentration, measured in terms of employment HHI (Columns 3, 4, 9 and 10); and the 50% of regions x occupations with the highest rate of job to job mobility (Columns 5, 6, 11 and 12).

B Data Appendix

B.1 Employment Stocks and Wages

The main data source used to study the effect of the reform on employment stocks and wages is the payroll tax registry called *DADS postes* (Déclarations des Données Sociales). This dataset records information on the universe of employment contracts signed within the French private sector in a given year. In particular, it records information on the number of hours worked in the year, the gross and net salary received, the occupation code, and some demographic characteristics of the worker including the age, gender, and whether the worker was born in France or abroad.⁴⁷ It further includes identifiers for the employing firm and establishment, which can be followed over time, and records their sector of activity and regional location. While this data provides a panel dimension at the firm and establishment level, workers' identifiers are not consistent from one year to the next, and therefore cannot be tracked. Nonetheless, each yearly database includes information on the situation of the individual in the previous year, including the employing firm identifier, occupation, wage and hours worked at $t-1$.

Most existing papers using worker panel data from France rely on the narrower *DADS panel*. The latter is provided by the French statistical office *INSEE* and consists in a sub-sample of 1/12th of the *DADS Postes*, selected based on workers' month of birth, that is complemented with individual identifiers consistent across years.⁴⁸ A recent paper by [Babet, Godechot and Palladino \(2022\)](#) shows that it is possible to use the overlapping information over two consecutive years to match workers between yearly files, based on the common information provided at t for the previous year and at $t-1$ for the following year, and to obtain a single match for 98% of the individuals. The matching cannot be established in the rare cases where several individuals have the exact same information, or where the information for a given individual was corrected from one wave to the next. Finally, individuals that go through career breaks that last more than one year cannot be connected, so are identified as different individuals if they reappear later on in the data. I follow the codes made available on the authors' website to construct individual worker identifiers that can be followed over time for the quasi-entirety of the *DADS postes*. Finally, [Babet, Godechot and Palladino \(2022\)](#) also document how in some years the country of birth variable is measured with noise, and provide a correction that assigns the foreign born status to all individuals that are

⁴⁷The information on the country of birth is deducted by the French statistics office *INSEE* based on the social security number, which starts with the two-digits code of the region of birth if the individual is born in France, and with 99 if born abroad.

⁴⁸Before 2002 the sample only included 1/24th of the entire workforce, and was increased to 1/12th from 2002 onward.

flagged as being born abroad at any point in time in the data. I follow their corrected indicator to assign the migrant status.

I then perform some additional cleaning on the data : I restrict the observations to individuals aged from 17 to 64 years old with consistent occupation codes, and that work between 1 and 360 days a year and between 1 and 14 hours per day. I drop contracts with gross wages below 6 euros per hour, which is about 1 euro below the minimum wage of the time, and I restrict the analysis to the French private sector without agriculture. Prior to 2008, only establishments with more than 20 employees are required to report the detailed occupation code of each employee,⁴⁹ while smaller establishments report it on a voluntary basis. To avoid potential biases linked to voluntary reporting, I exclude from the sample plants with less than 20 employees. In order to avoid capturing endogenous changes of sector and/or location, I assign these attributes based on the first year of appearance of each establishment, and keep them fixed over time. Finally, to avoid capturing endogenous reclassification of the occupation of individuals within firms, I fix them based on the first occupation reported by the individual when joining the plant.

The final dataset used in the analysis is at the level of establishments $\times$ occupations $\times$ years. Employment is thus summed within cells, while the hourly wages are averaged within cells, separately for natives and migrants. The main outcomes of interests are the following :

- Total employment stock of migrants and natives in the establishment $\times$ occupation, expressed in terms of full-time equivalent.
- Average wages of migrants and natives in the establishment $\times$ occupation.
- Average wages of incumbent migrants and natives in the establishment $\times$ occupation, defined as the wages of individuals that were already working in France before 2008.
- Average wages of migrants and natives in the establishment $\times$ occupation, residualized of individual characteristics including age, age squared, gender, and years of experience within the firm.⁵⁰

The average wage of migrants and natives captures the evolution of the overall conditions in the occupation for each type of worker. The average wages of incumbent workers captures the effect of the reform on the occupational conditions net of composition changes introduced by

⁴⁹Occupation codes are reported at the most disaggregated level of the French PCS classification: 4-digits.

⁵⁰To obtain the residualized wage information, I estimate an individual level regression with log wage on the left-hand side and demographic characteristics on the right-hand side, including age, age squared, years of experience in the firm, and a gender dummy. The regression is estimated separately for native and migrant workers and the residuals are then averaged within the establishment - occupation cells.

the newly arrived workers, but also excludes the effect on young workers that just entered the labor force. Finally, the average wage residualized of individual characteristics aims at correcting for compositional effects without excluding from the sample recent graduates. In the analysis, I present the results obtained on each of these outcomes.

The period considered in the regression analysis goes from 2004 to 2010, including 4 years preceding the reform and 3 years of post-reform effects. The starting year is determined by the fact that the data initially made available for this study started in 2004, and in the interest to avoid the pre-EU enlargement period, which took place in 2004, and the change in the occupation classification, introduced in 2002 but fully deployed only by 2003. The end year is chosen because in 2011 the list was temporarily restricted to roughly half of the original occupations, and was then re-set to its initial composition in 2012.⁵¹ Nonetheless, in the event study graphs the post-trends are shown up to 2013, to give a sense of how the effect evolved in the medium term.

B.2 Hiring flows

To test the robustness of the effect on employment I also look at the impact of the reform on hiring flows measured using a different administrative database. The data on hiring flows used for this analysis is derived from two sources: the monthly Declaration of Labor Movements (Déclarations des Mouvements de Main d'Oeuvre - DMMO), which is collected for all establishments with more than 50 employees, and the quarterly Survey of Labor Movements (Enquête sur les mouvements de main d'oeuvre - EMMO), which is collected for a representative sample of establishments with less than 50 employees. The scope of the data includes all the private sector within the Metropolitan French territory, except for agriculture and temporary agencies of employment. For each flow taking place within an establishment, the data reports the detailed occupation concerned,⁵² an indicator for whether it is an entry or an exit, an indicator for whether it is a short term contract or an open ended one, and information on whether exit flows are voluntary (e.g. quits and retirements), or involuntary (e.g. lay-offs and non-renewals of short term contracts). At the individual level, the data reveals the gender and the age of the worker, as well as its broad country of birth group. At the establishment level, the data reports the number of employees, the sector and the region where the plant is operating. Establishments can be followed over time but not workers. Here as well I fix sector and region according to the one of first appearance of the establishment.

⁵¹In 2011 the government tried to reduce the list with the claim that, given the recession, the priority for jobs should be given to French nationals and current residents. However, due to irregularities in the procedure, the change was invalidated one year after, and the list went back to its original composition up to this day.

⁵²Occupation codes are reported at the most disaggregated level of the French PCS classification: 4-digits.

The final dataset used for the analysis is at the level of establishments $\times$ occupations $\times$ years. Entries and exits are thus summed within cells, separately for French natives and for migrants. The main outcomes of interests are the number of entries, the probability of hiring (dummy equals to 1 if entries are bigger than 0), and the probability of net positive entry (dummy equals to 1 if the number of entries exceeds the number of exits).

B.3 Treatment and control occupations

To identify treated and control occupations, I merge to the main datasets the list of occupations of the reform, and the tightness indicators produced by the French employment office. Both of these sources rely on a different occupation classification with respect to the *DADS postes*: they use the ROME classification instead of the PCS classification. The French statistical office attached to the Ministry of Labor (DARES) provides a crosswalk table to transpose the ROME codes into the PCS codes.⁵³

The commission in charge of defining which occupations to include in the list of the reform considered seven different tightness criteria measured at the level of occupation $\times$ region : i) the ratio between the number of vacancies posted and the number of registered unemployed workers qualified for the job (commonly known as the tightness indicator), ii) the volume of vacancies, iii) the volume of job seekers, iv) the evolution of the stock of vacancies, v) the evolution of the stock of job seekers, vi) the turnover rate of job seekers at the end of the month, and vii) the share of long-term contracts within vacancies. In order to obtain a single measure of tightness combining all of these criteria, I translate each element into a standardized z-score using the formula $Zscore_i = \frac{x_i - \bar{X}}{\sigma_X}$, and obtain the final index by averaging the seven z-scores.⁵⁴ The latter captures all the information used by the commission to define the list of occupations for the reform.

⁵³<https://dares.travail-emploi.gouv.fr/donnees/la-nomenclature-des-familles-professionnelles-fap-2009>

⁵⁴I flip the sign of the z-scores for the number of job seekers and their evolution before computing the final index, such that each measure consistently presents higher values for greater tightness.

C A model of labor demand with monopsonistic competition

The production function in the model proposed by [Amior and Manning \(2020\)](#) follows to the one proposed by [Card \(2009\)](#); [Manacorda, Manning and Wadsworth \(2012\)](#); [Ottaviano and Peri \(2012\)](#). The output Y_t of a representative firm is produced using labor inputs from different skill groups indexed by o , which are assembled through a CES production function:

$$Y_t = A_t \left[\sum_o \theta_{ot} \mathbf{L}_{ot}^{\frac{\omega-1}{\omega}} \right]^{\frac{\omega}{\omega-1}} \quad (10)$$

where ω is the elasticity of substitution between the skill groups and θ_{ot} is the skill group-specific efficiency parameter that might vary through time. Without loss of generality, θ_{ot} is normalized such that $\sum_o \theta_{ot} = 1$. For simplicity, capital is assumed to be fixed, and it is not reported. Further, the labor in each skill group L_{ot} combines inputs from native workers N_{ot} and migrant workers M_{ot} through a CES production function nested into the first level:

$$\mathbf{L}_{ot} = \left[\beta_{ot} M_{ot}^{\frac{\sigma-1}{\sigma}} + N_{ot}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (11)$$

where σ is the elasticity of substitution between natives and migrants, and β_{ot} is the relative efficiency of migrants with respect to natives within each skill group, which can vary through time if, for instance, there is a change in the relative quality of migrants or in discrimination biases. The existence of imperfect substitutability between migrants and natives within cells can be rationalized by the fact that each skill group is an aggregation of unobservable tasks in which the distribution might differ between natives and migrants due to differences in comparative advantage. Differently from the model of [Ottaviano and Peri \(2012\)](#), here, each firm faces a labor supply curve that is upward sloping in wage, as in [Card et al. \(2018\)](#):

$$Q = Q_0 (W - R_Q)^{\epsilon_Q} \quad (12)$$

Where $Q = \{N, M\}$ within each skill group. The labor supply of natives and migrants depend on the available stock Q_0 (a function of the wage offered by other firms), the wage offered by the firm W , the worker specific reservation wage R_Q and an elasticity parameter ϵ_Q , which captures idiosyncratic preferences but also job search efficiency. In the context of this paper, there are

strong reasons to believe that $\epsilon_M < \epsilon_N$, given that migrants are subject to legal restrictions on their ability to search for different jobs. Under these circumstances, the marginal cost function for each worker type is the following:

$$MC_Q(W) = W + \frac{W - R_Q}{\epsilon_Q} \quad (13)$$

The log-linearized wage equations of migrant and native workers can be derived as follows:

$$\ln(w_{ot}^M) = \ln(A_t) + \frac{1}{\omega} \ln(Y_t) + \ln(\theta_{ot}) + \ln(\beta_{ot}) - \frac{1}{\sigma} \ln(M_{ot}) + \left(\frac{1}{\omega} - \frac{1}{\sigma}\right) \ln(L_{ot}) - \phi_{ot}^M \left(\frac{M_{ot}}{N_{ot}}\right) \quad (14)$$

$$\ln(w_{ot}^N) = \ln(A_t) + \frac{1}{\omega} \ln(Y_t) + \ln(\theta_{ot}) - \frac{1}{\sigma} \ln(N_{ot}) + \left(\frac{1}{\omega} - \frac{1}{\sigma}\right) \ln(L_{ot}) - \phi_{ot}^N \left(\frac{M_{ot}}{N_{ot}}\right) \quad (15)$$

These formulas equal wages to the marginal product of migrant and native labor, minus their specific mark-down (ϕ_{ot}^M and ϕ_{ot}^N). The mark-down is decreasing in both the reservation wage R_Q and in the supply elasticity ϵ_Q .⁵⁵ Mark-downs may also depend on the cell-level migrant share, if at least some firms cannot perfectly discriminate the wages paid to migrants and natives. Specifically, the authors assume that ϕ_{ot}^M and ϕ_{ot}^N can be approximated by log-linear functions of $\frac{M}{N}$ as follows:

$$\phi_{ot}^N \left(\frac{M_{ot}}{N_{ot}}\right) = \phi_{ot,0}^N + \phi_{ot,1}^N \ln \left(\frac{M_{ot}}{N_{ot}}\right) \quad (16)$$

$$\phi_{ot}^M \left(\frac{M_{ot}}{N_{ot}}\right) = \phi_{ot,0}^M + \Delta\phi_{ot,0} + (\phi_{ot,1}^N + \Delta\phi_{ot,1}) \ln \left(\frac{M_{ot}}{N_{ot}}\right) \quad (17)$$

where $\Delta\phi_{ot,0}$ is the difference in mark-down between migrants and natives that does not depend on the migrant share, and $\Delta\phi_{ot,1}$ is the difference in mark-down between migrants and natives that does depend on the migrant share and thus might be impacted by the reform.⁵⁶ This allows

⁵⁵Amior and Manning (2020) show that $\phi^Q = \ln\left(\frac{\epsilon_Q + 1}{\epsilon_Q + \frac{R_Q}{MP}}\right)$, under the assumption that firms can pay different wages to migrants and natives.

⁵⁶This empirical analysis assumes that the reform only impacted $\Delta\phi_{ot,1}$ through a change in the migrant share. Given that the reform made it slightly easier for migrants to change employers, as shown in Section 6.1, it might also have slightly decreased $\Delta\phi_{ot,1}$ through this channel, but we consider the latter effect to be second order relative to the main one.

to express the relative wage of migrants to natives using the following formula:

$$\ln\left(\frac{w_{ot}^M}{w_{ot}^N}\right) = \ln(\beta_{ot}) - \Delta\phi_{ot,0} - \left(\frac{1}{\sigma} + \Delta\phi_{ot,1}\right)\ln\left(\frac{M_{ot}}{N_{ot}}\right) \quad (18)$$

The canonical model of [Borjas \(1999\)](#) assumes perfect competition ($\Delta\phi_{ot,0}=\Delta\phi_{ot,1}=0$) and perfect substitution in production between migrants and natives within the same skill group ($\sigma = \infty$), such that changes in their relative supply have no impact on their relative wages. [Ottaviano and Peri \(2012\)](#) relax the assumption of perfect degree of substitution but maintain the assumption of perfect competition, which allows them to derive the elasticity of substitution between migrants and natives by estimating equation 6 empirically. [Amior and Manning \(2020\)](#) reveal that, once the assumption of equal mark-downs is relaxed, it is not possible anymore to disentangle the elasticity of substitution parameter (σ) from the differential mark-down parameter that depends on the migrant share ($\Delta\phi_{ot,1}$).